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Applied Programming APPR6312
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PART 2

Section A: Extending the Web App with Azure Functions

A.1: Function Project inside the existing solution.

Figure 1: Gift of the Givers Disaster Relief System solution showing the newly added GiftOfGiversFunctions Azure Functions project with an HTTP Trigger function.

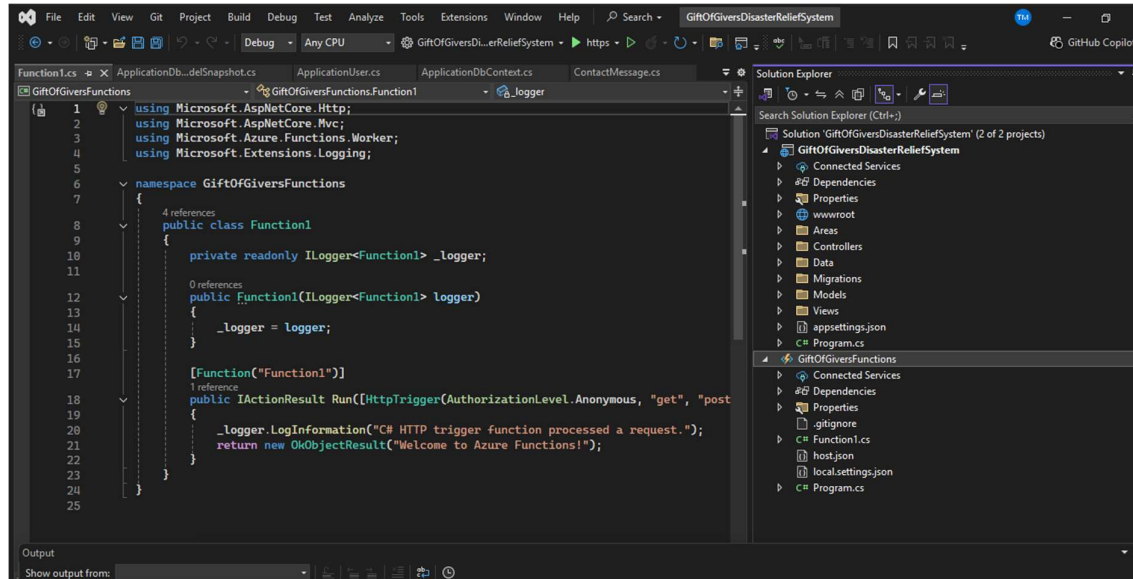
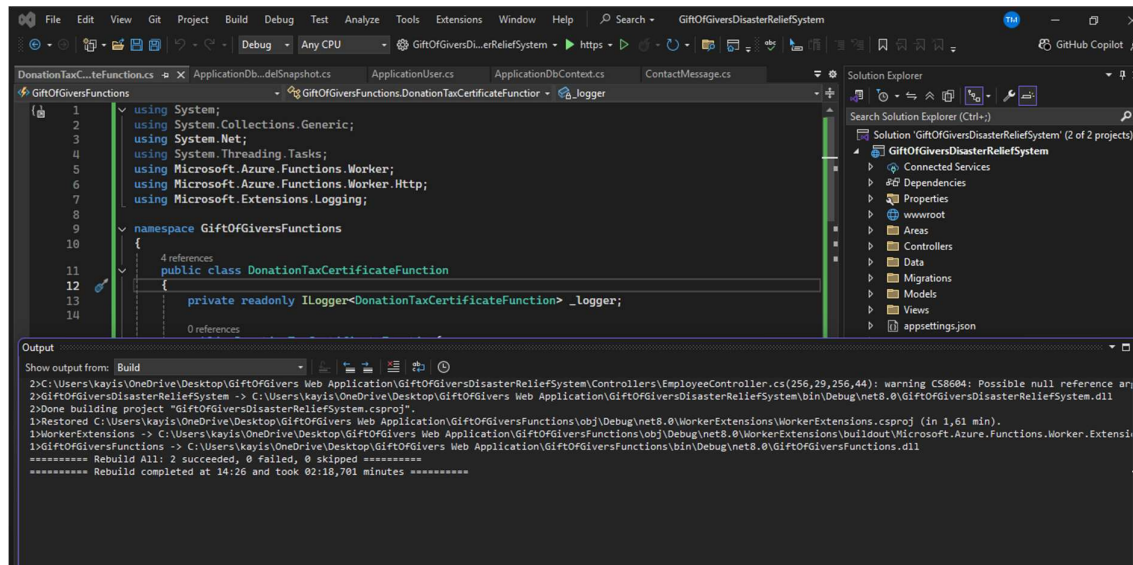


Figure 2: Successful rebuild of the Gift of the Givers solution showing that both the web application and Azure Functions projects built successfully with zero failed projects.



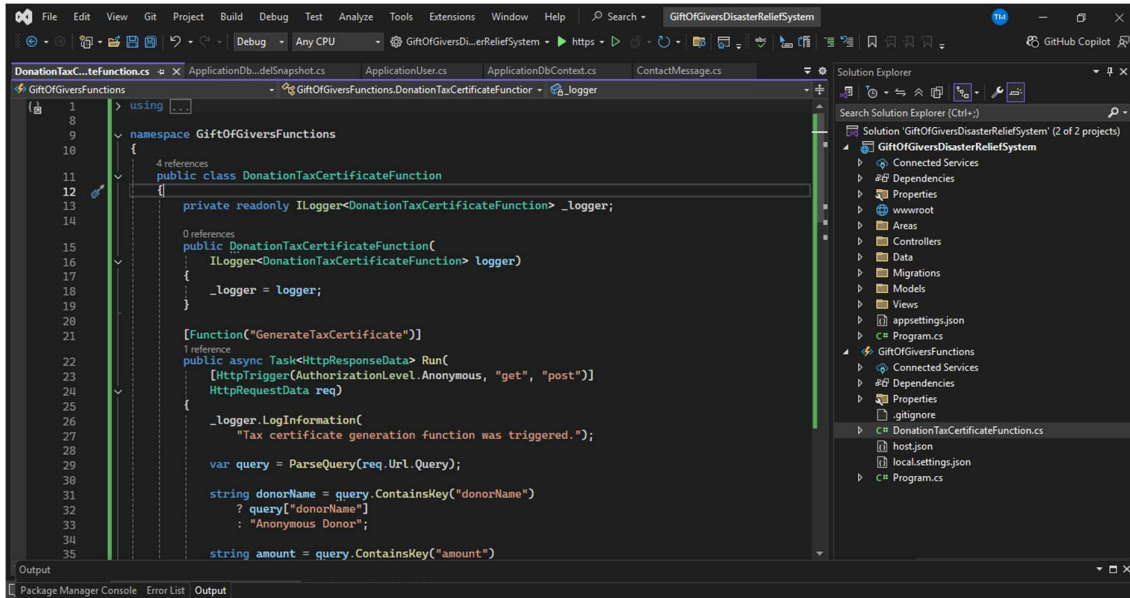


Figure 3: Successful rebuild of the Gift of the Givers web application and Azure Functions projects after clearing the build folders.

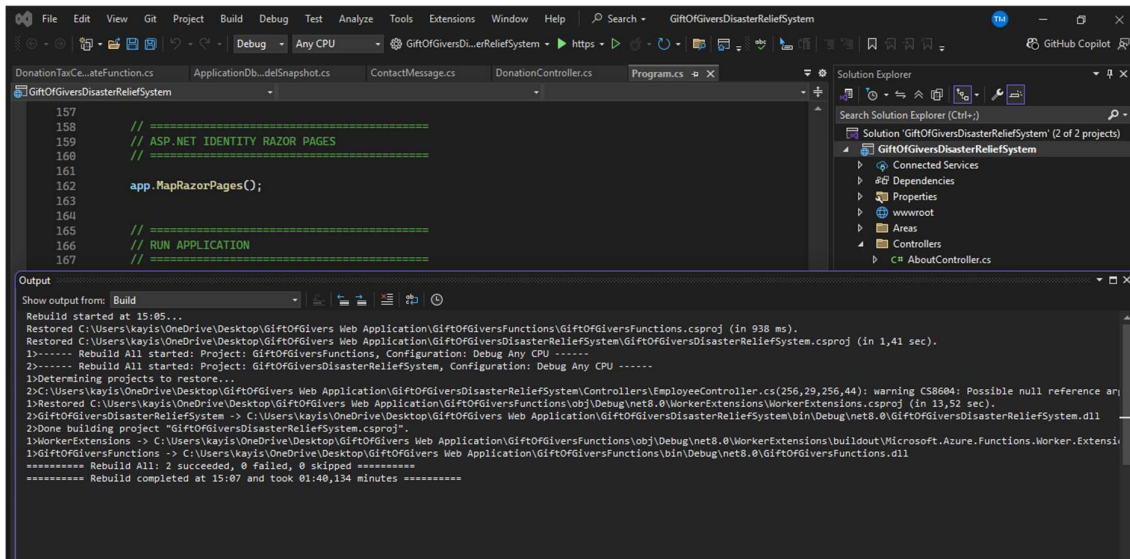


Figure 4: Updated Program.cs showing the registration of HttpClient for communication between the web application and Azure Functions.

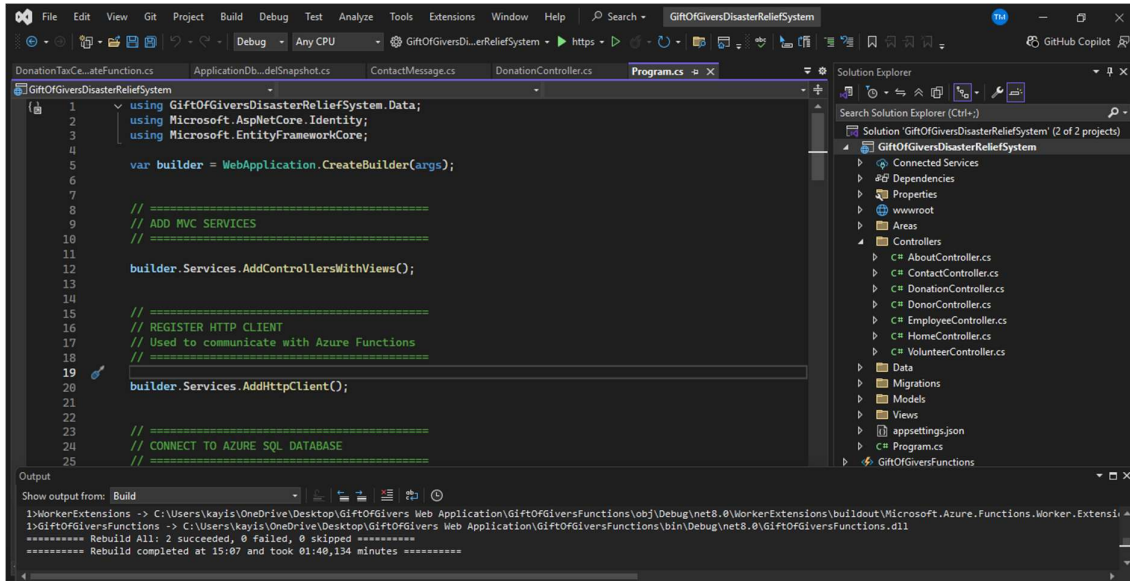


Figure 5: Successful rebuild of the Gift of the Givers web application and Azure Functions projects after clearing the Function build files.

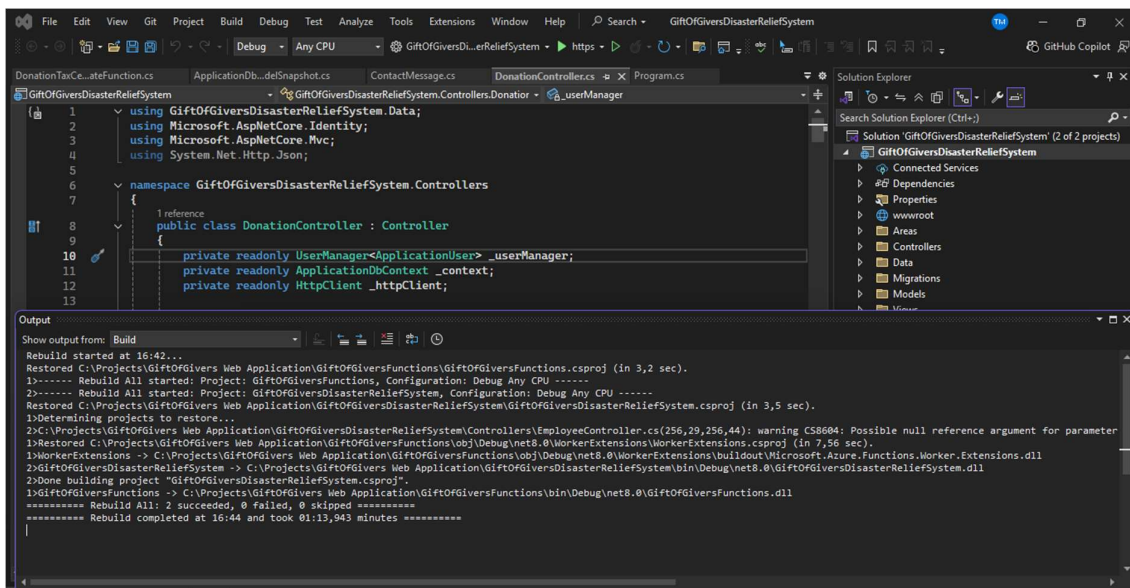
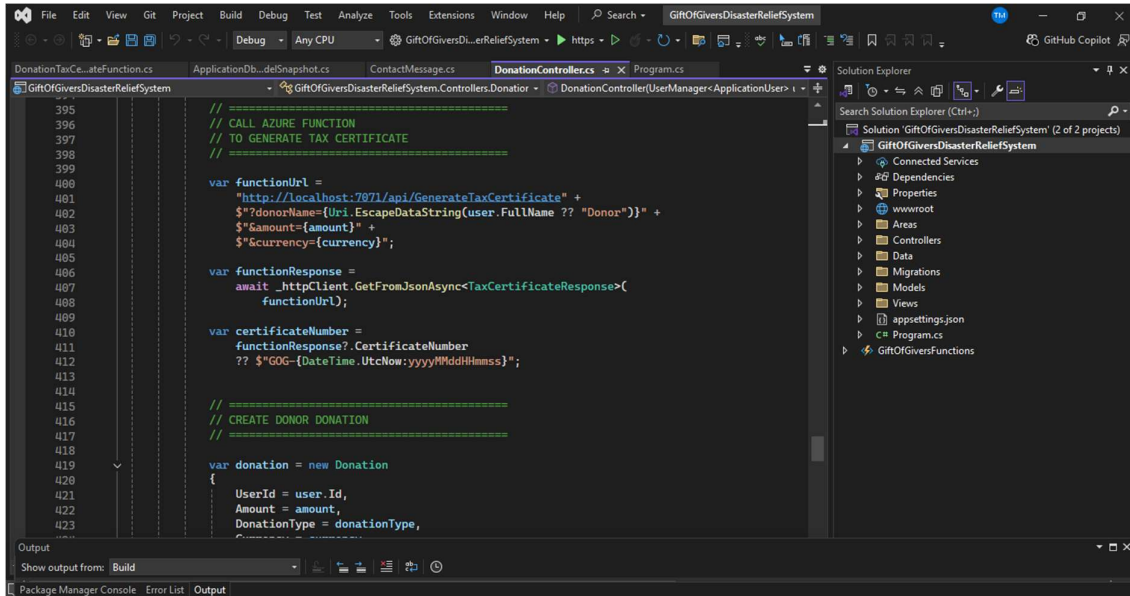


Figure 6: DonationController calling the GenerateTaxCertificate Azure Function to generate a dummy tax certificate for a donor.



A.2 Run and Test the Function Locally

Figure 7: Azure Functions Core Tools running locally and displaying the GenerateTaxCertificate HTTP trigger on localhost:7033.

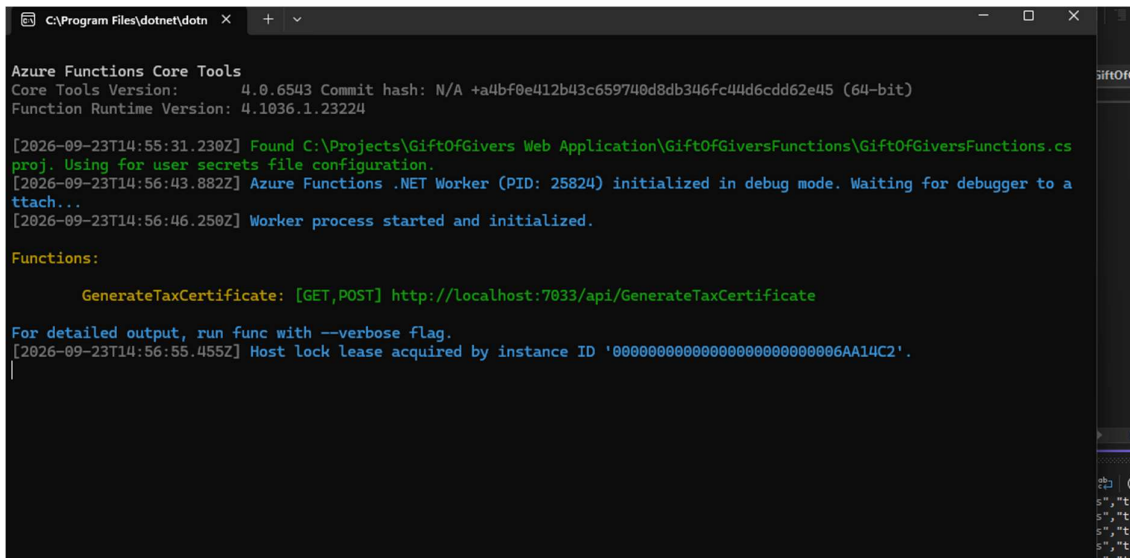


Figure 8: Browser test of the GenerateTaxCertificate Azure Function returning a dummy tax certificate successfully.

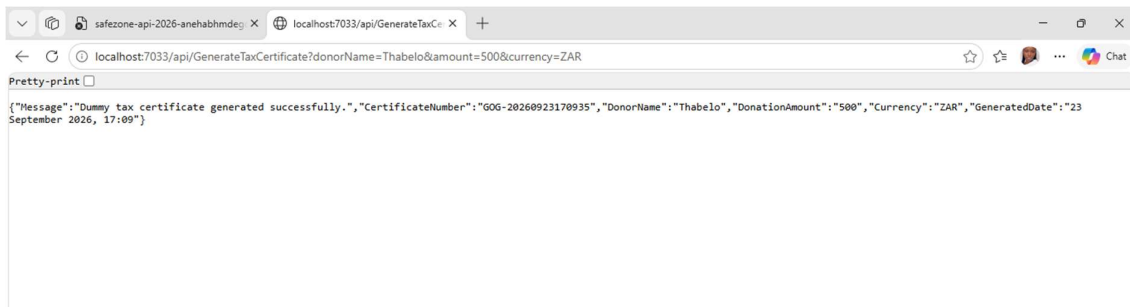


Figure 9: Postman test of the GenerateTaxCertificate Azure Function successfully returning a dummy tax certificate.

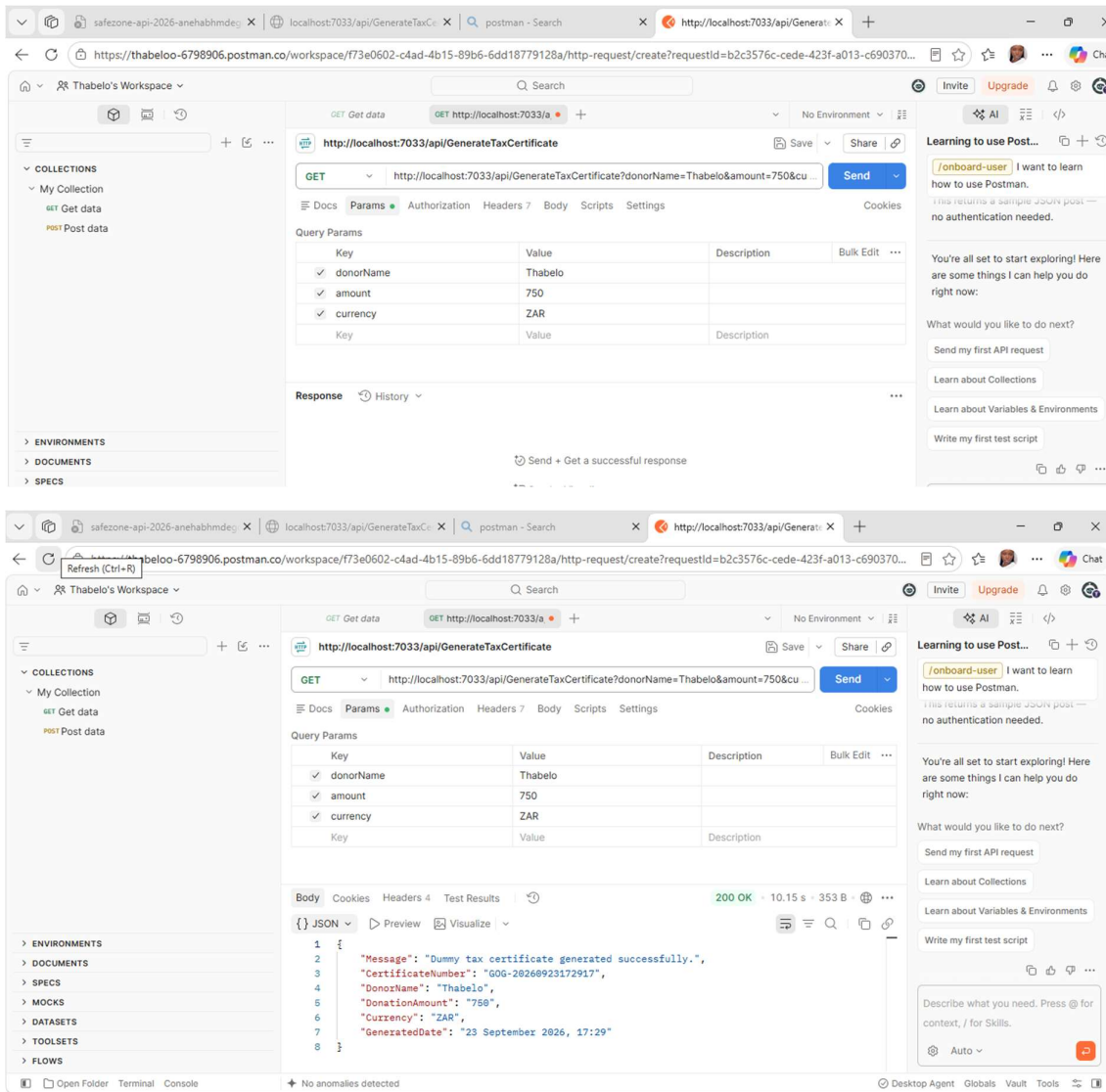


Figure 10: Successful rebuild after adding Function validation and error handling.

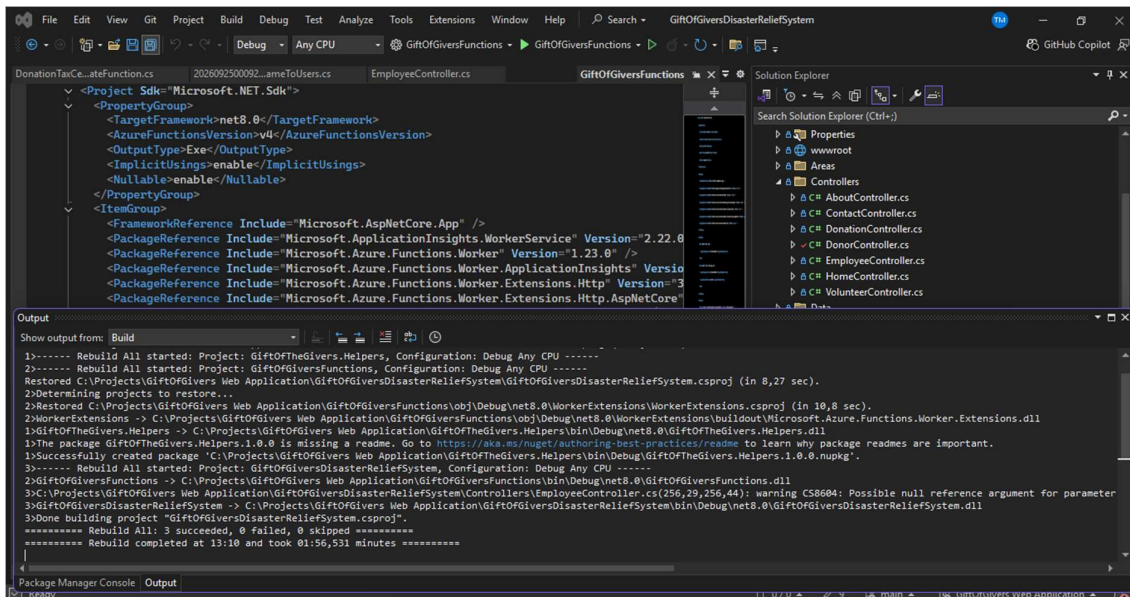


Figure 11: Azure Function running locally with the GenerateTaxCertificate HTTP trigger.

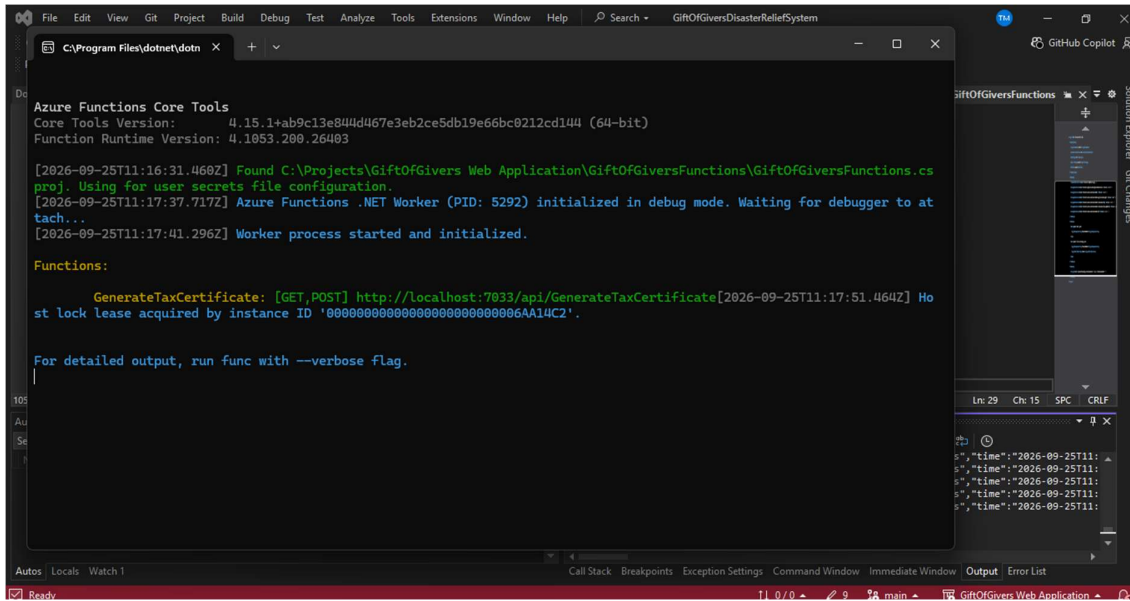


Figure 12: Successful local test of the GenerateTaxCertificate function with valid donation parameters.

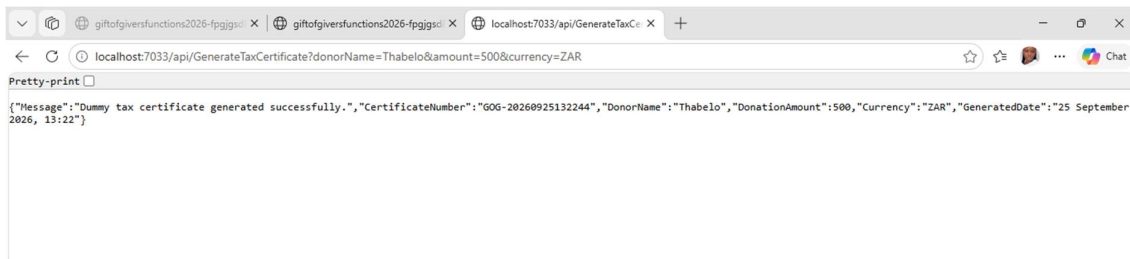
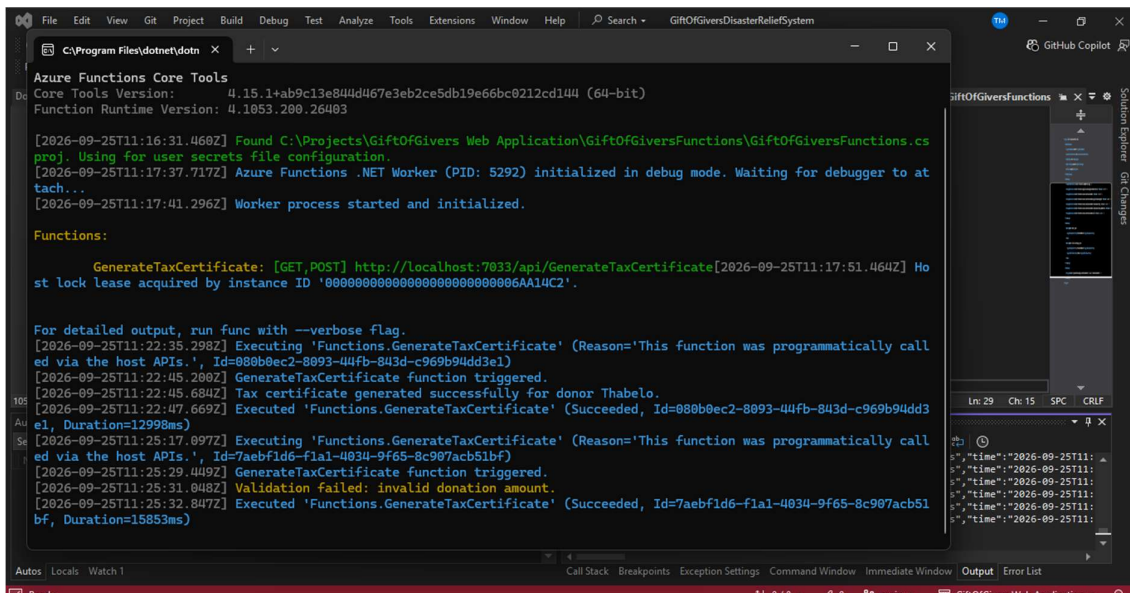
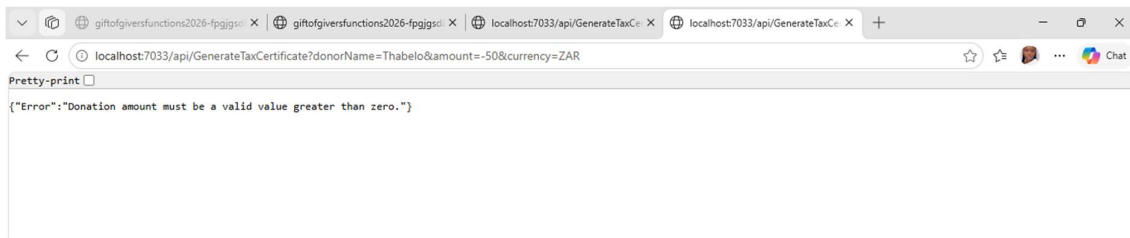


Figure 13: Local Function test showing validation of an invalid donation amount.



A.3 Deploy the Function to Azure

Figure 14: Successful Azure Function App deployment

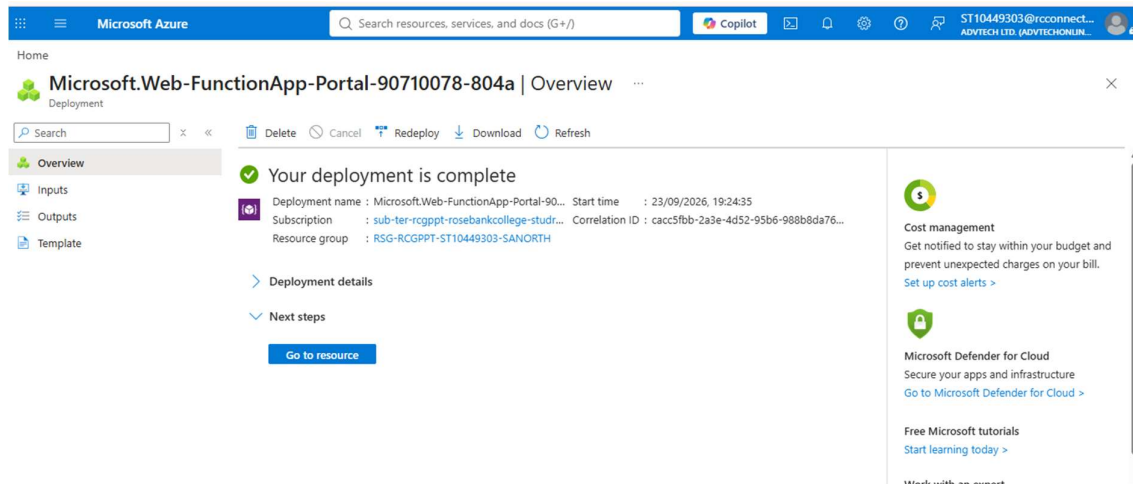


Figure 15: Azure Function App overview showing the running giftogiversfunctions2026 Function App

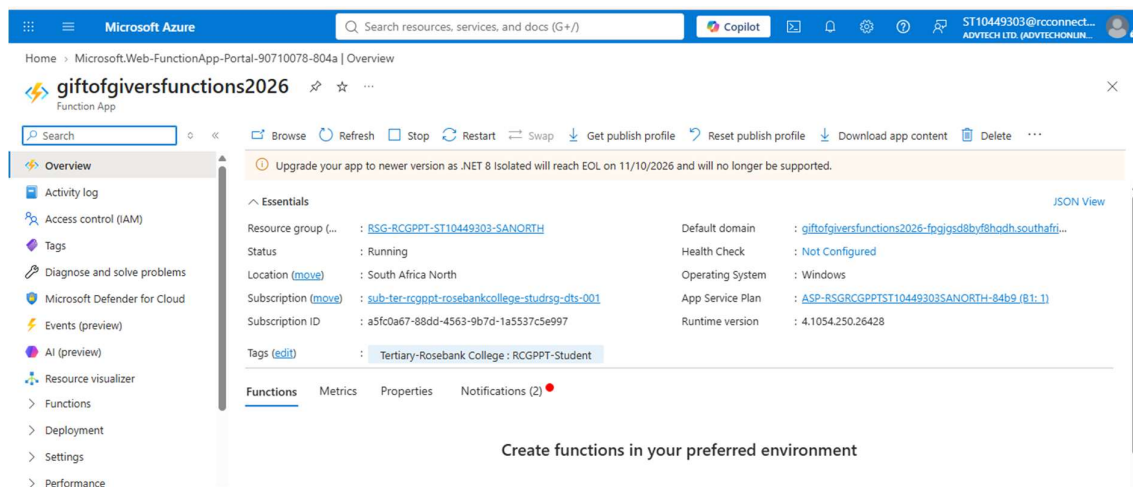


Figure 16: Visual Studio Publish profile configured for the giftogiversfunctions2026 Azure Function App

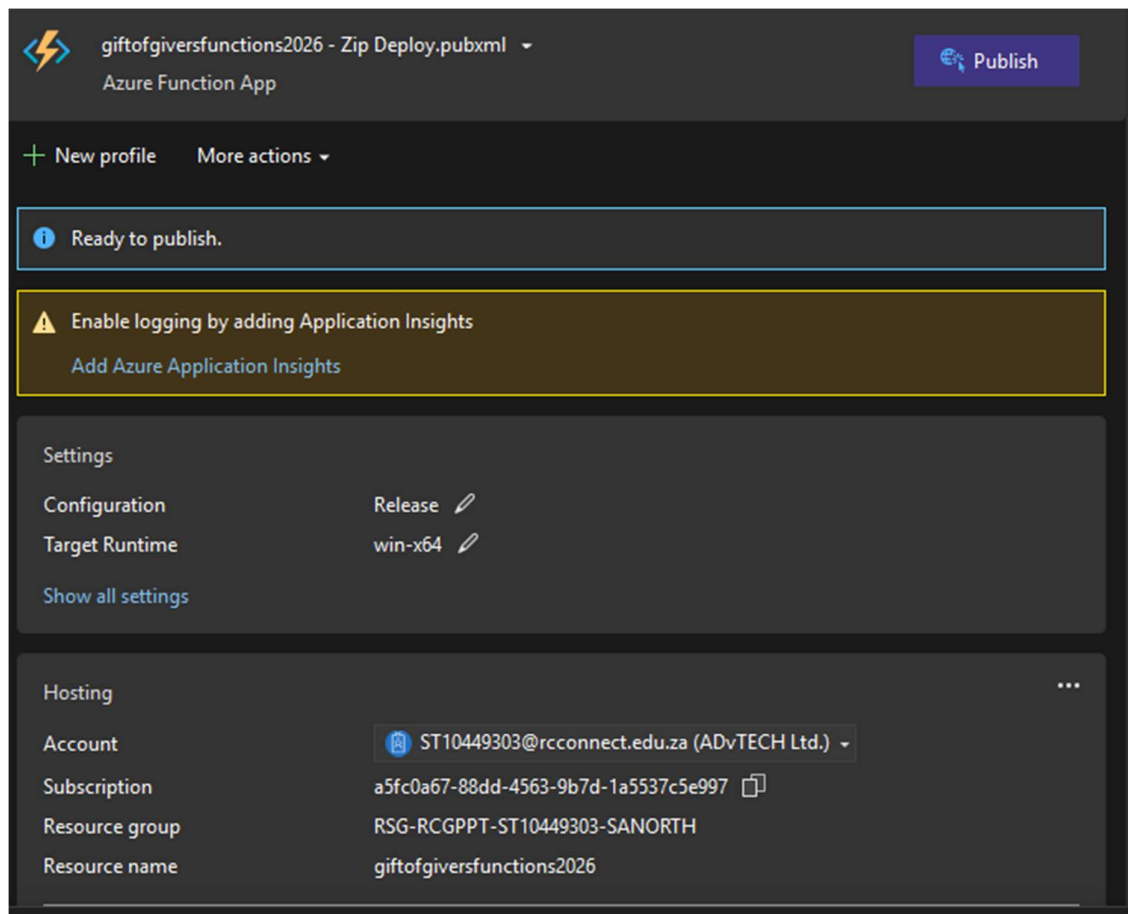


Figure 17: Successful publication of the GiftOfGiversFunctions project to Azure

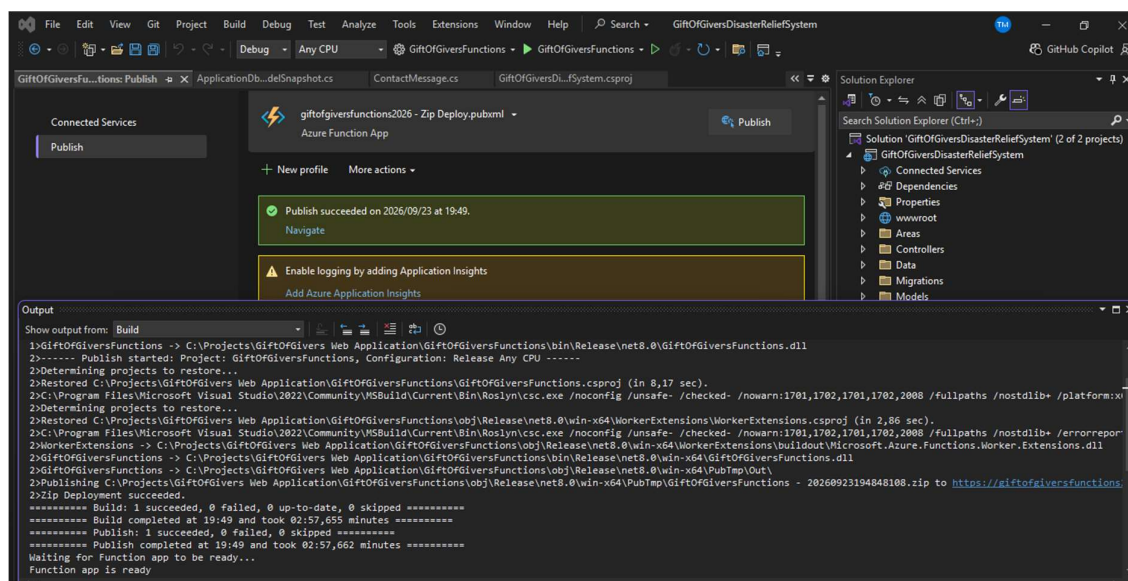
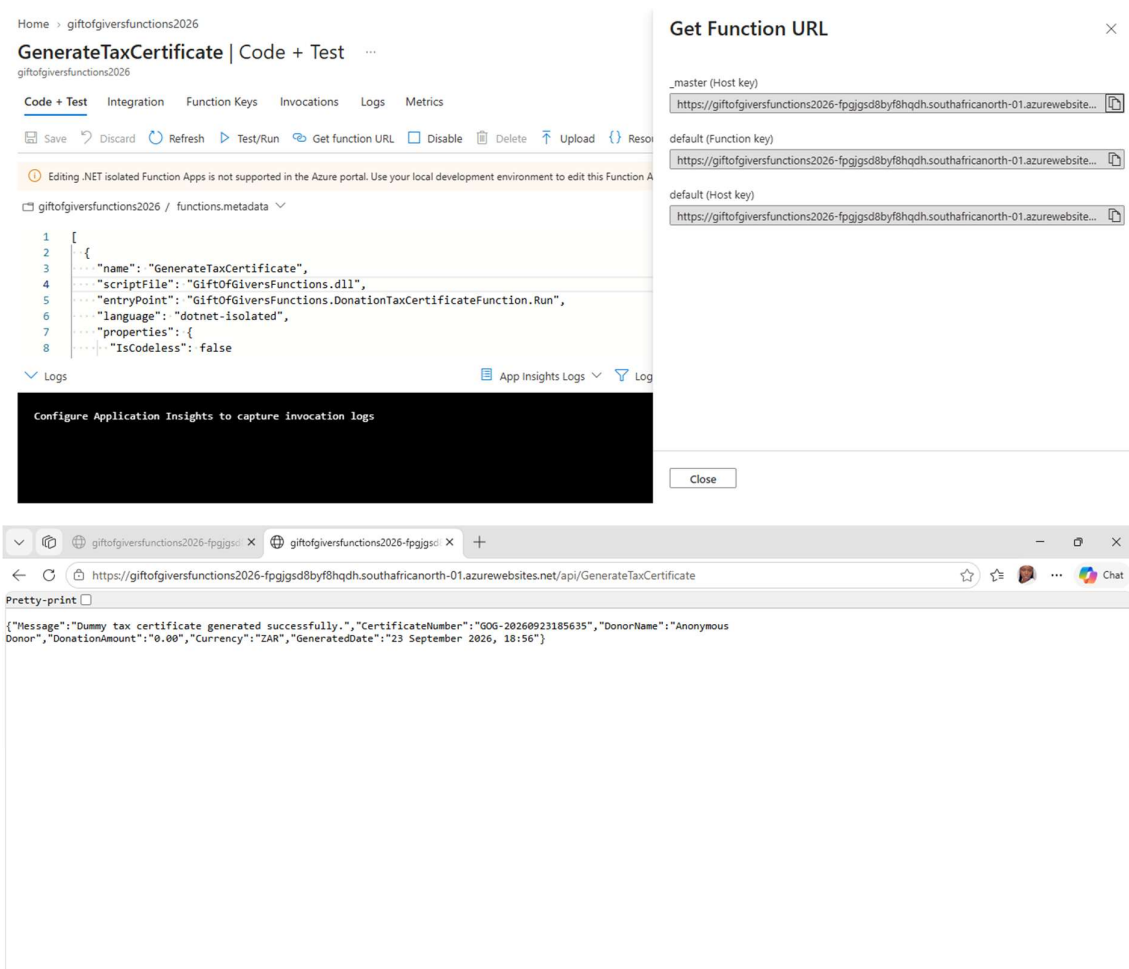


Figure 18: Azure Portal displaying the live Function URL for the GenerateTaxCertificate function.

Live Function URL: <https://giftofgiversfunctions2026-fpgjgsd8byf8hqdh.southafricanorth-01.azurewebsites.net/api/GenerateTaxCertificate>

<https://giftofgiversfunctions2026-fpgjgsd8byf8hqdh.southafricanorth-01.azurewebsites.net/api/GenerateTaxCertificate>



Section B: Managing Code with Azure Repos Git

B.1 Connect Visual Studio Project to Azure Repos

Figure 19: Empty Azure Repos Git repository created for the Gift Of The Givers Disaster Relief System

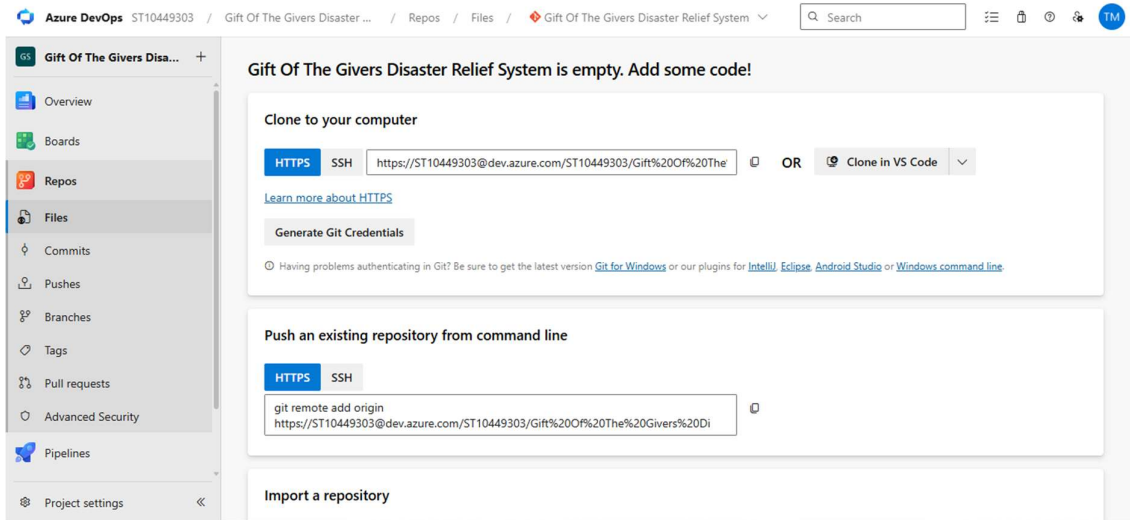


Figure 20: Azure DevOps repository showing the pushed Gift of the Givers web application and Azure Functions project.

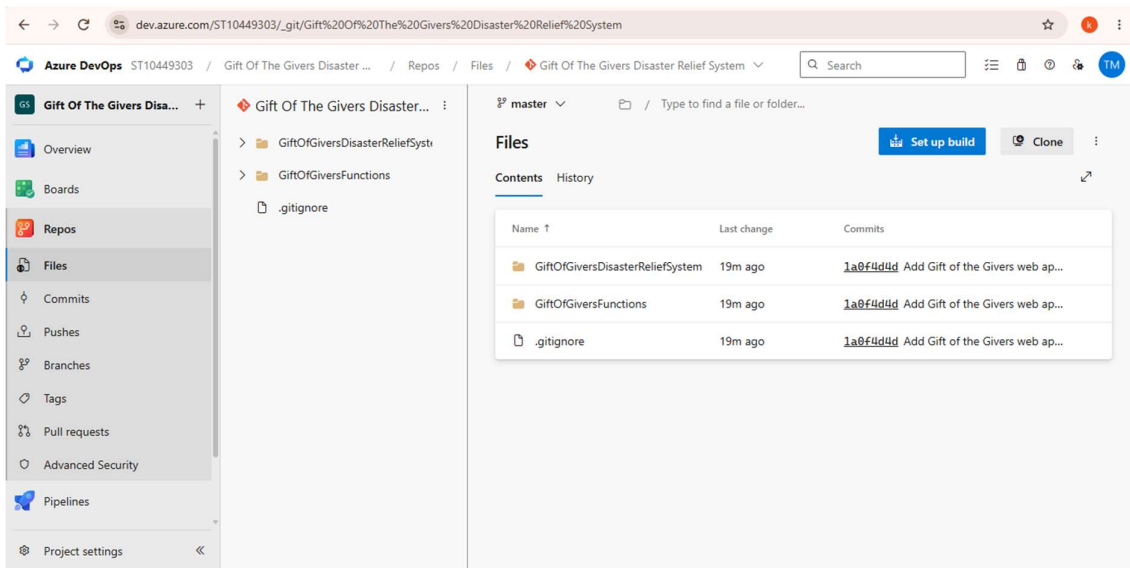
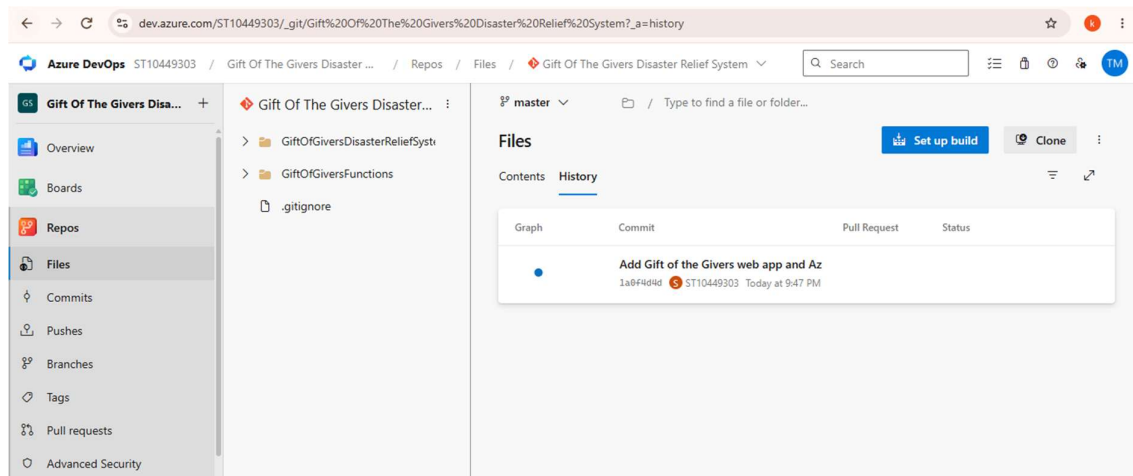
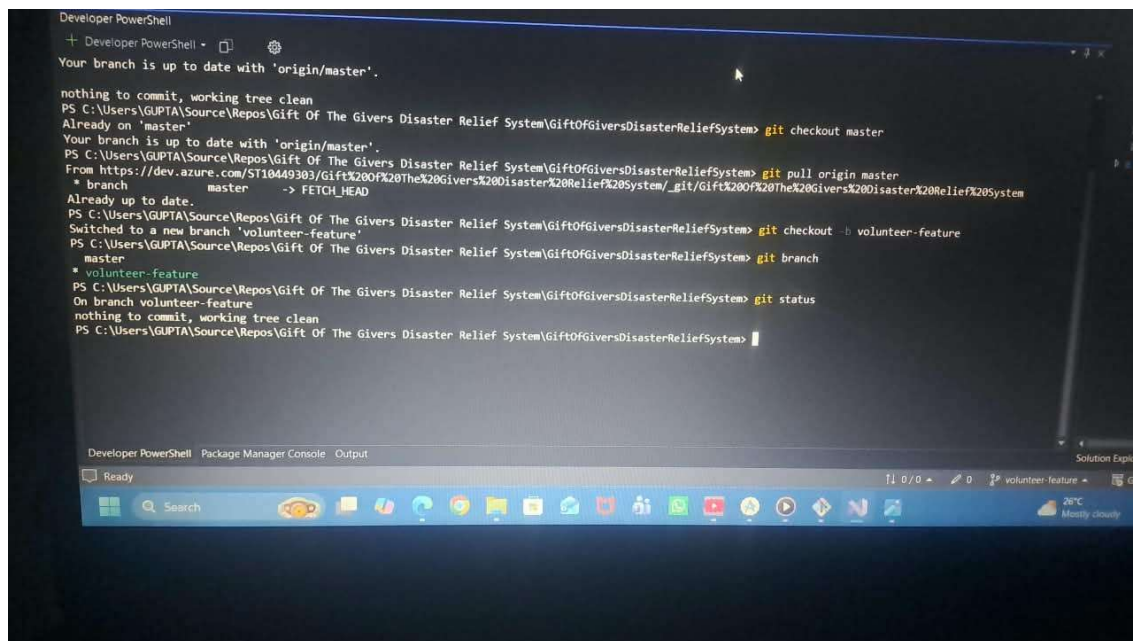


Figure 21: Azure DevOps commit history showing the initial project commit.



B.2 Branching & Merging in Visual Studio

Figure 22: Visual Studio Terminal showing the volunteer-feature branch created from master.




```
no changes added to commit (use "git add" and/or "git commit -a")
PS C:\Users\GUPTA\Source\Repos\Gift Of The Givers Disaster Relief System\GiftOfGiversDisasterReliefSystem> git add .
PS C:\Users\GUPTA\Source\Repos\Gift Of The Givers Disaster Relief System\GiftOfGiversDisasterReliefSystem> git commit -m "Update volunteer feature"
[volunteer-feature 4fa7b2d] Update volunteer feature
1 file changed, 1 insertion(+)
PS C:\Users\GUPTA\Source\Repos\Gift Of The Givers Disaster Relief System\GiftOfGiversDisasterReliefSystem> git push
Enumerating objects: 9, done.
Counting objects: 100% (9/9), done.
Delta compression using up to 8 threads
Compressing objects: 100% (5/5), done.
Writing objects: 100% (5/5), 483 bytes | 483.00 KiB/s, done.
Total 5 (delta 4), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Analyzing objects... (5/5) (10 ms)
remote: Validating commits... (1/1) done (0 ms)
remote: Storing packfile... done (51 ms)
remote: Storing index... done (35 ms)
remote: Updating refs... done (136 ms)
To https://dev.azure.com/ST10449303/GiftOfTheGiversDisasterReliefSystem/_git/GiftOfTheGiversDisasterReliefSystem
1a0f4d4..4fa7b2d volunteer-feature -> volunteer-feature
PS C:\Users\GUPTA\Source\Repos\Gift Of The Givers Disaster Relief System\GiftOfGiversDisasterReliefSystem>
```

Figure 23: Successful commit and push of the volunteer-feature branch to Azure Repos.

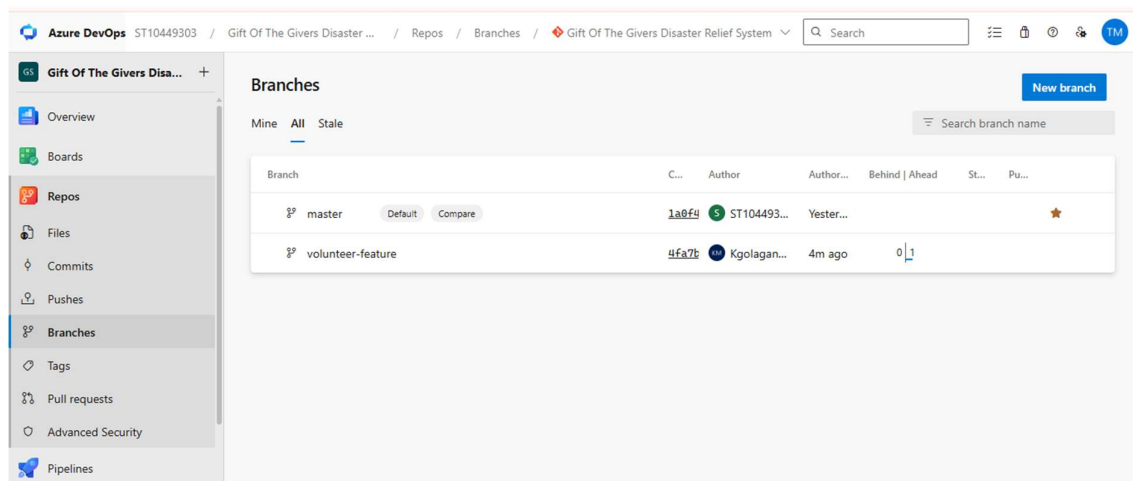


Figure 24: Visual Studio Terminal showing the contact-feature branch active.

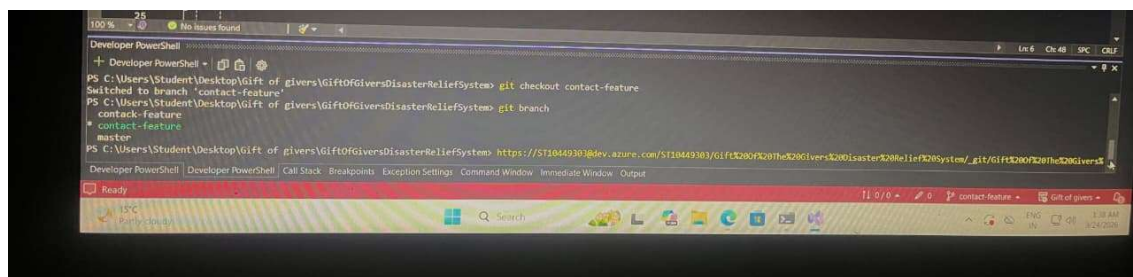


Figure 25: Contact feature change detected on the contact-feature branch.

The screenshot shows the Visual Studio IDE with the 'ContactController.cs' file open. The code defines a `ContactController` class with a constructor that takes an `ApplicationDbContext` and a `Context` parameter. A comment above the constructor indicates a change for the Azure Repos branching task. The 'Solution Explorer' on the right shows the project structure, including controllers, data, migrations, models, and views. The 'Developer PowerShell' at the bottom displays the output of the `git status` command, showing that the `contact-feature` branch is active and that the `Controllers/ContactController.cs` file has been modified.

```
private readonly ApplicationDbContext _context;

0 references | 5710449303, 11 hours ago | 1 author, 1 change
public ContactController(ApplicationDbContext context)
{
    _context = context;
}

// Contact feature updated for Azure Repos branching task
// CONTACT PAGE
// =====
```

```
** Visual Studio 2022 Developer PowerShell v17.14.25
** Copyright (c) 2025 Microsoft Corporation
*****
PS C:\Users\Student\Desktop\Gift of givers\GiftOfGiversDisasterReliefSystem> git status
On branch contact-feature
Changes not staged for commit:
  (use "git add <file>..." to update what will be committed)
  (use "git restore <file>..." to discard changes in working directory)
        modified:   Controllers/ContactController.cs

no changes added to commit (use "git add" and/or "git commit -a")
PS C:\Users\Student\Desktop\Gift of givers\GiftOfGiversDisasterReliefSystem>
```

Figure 26: Successful commit and push of the contact-feature branch to Azure Repos.

The screenshot shows the Visual Studio IDE with the 'ContactController.cs' file open. The 'Developer PowerShell' at the bottom displays the output of the `git add`, `git commit`, and `git push` commands. The `git add` command shows that the `Controllers/ContactController.cs` file has been staged. The `git commit` command shows that the commit message is 'Update contact feature'. The `git push` command shows that the commit has been pushed to the `contact-feature` branch on the remote repository. The output also includes a link to create a pull request on Azure DevOps.

```
(use "git restore --staged <file>..." to unstage)
modified:   Controllers/ContactController.cs

PS C:\Users\Student\Desktop\Gift of givers\GiftOfGiversDisasterReliefSystem> git add
Nothing specified, nothing added.
hint: Maybe you wanted to say "git add ."?
hint: Disable this message with "git config set advice.addemptyPaths false"
PS C:\Users\Student\Desktop\Gift of givers\GiftOfGiversDisasterReliefSystem> git commit -m "Update contact feature"
[contact-feature 5710449303] Update contact feature
1 file changed, 1 insertion(+), 1 deletion(-)
PS C:\Users\Student\Desktop\Gift of givers\GiftOfGiversDisasterReliefSystem> git push -u origin contact-feature
Enumerating objects: 0, done.
Counting objects: 100% (0/0), done.
Delta compression using up to 4 threads
Compressing objects: 100% (5/5), done.
Writing objects: 100% (5/5), 409 bytes | 249.00 KiB/s, done.
Total 5 (delta 4), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Analyzing objects... (5/5) (8 ms)
remote: Validating commits... (1/1) done (0 ms)
remote: Storing packfile... done (38 ms)
remote: Storing index... done (36 ms)
remote: Updating refs... done (154 ms)
remote:
remote: Create a pull request for 'contact-feature' by visiting:
remote:   https://dev.azure.com/5710449303/GiftOfGiversDisasterReliefSystem/_git/GiftOfGiversDisasterReliefSystem/pull
remote:   requests/create?sourceRef=contact-feature
remote:
remote: To https://dev.azure.com/5710449303/GiftOfGiversDisasterReliefSystem/_git/GiftOfGiversDisasterReliefSystem:
[ new branch ]
branch 'contact-feature' set up to track 'origin/contact-feature'.
PS C:\Users\Student\Desktop\Gift of givers\GiftOfGiversDisasterReliefSystem>
```

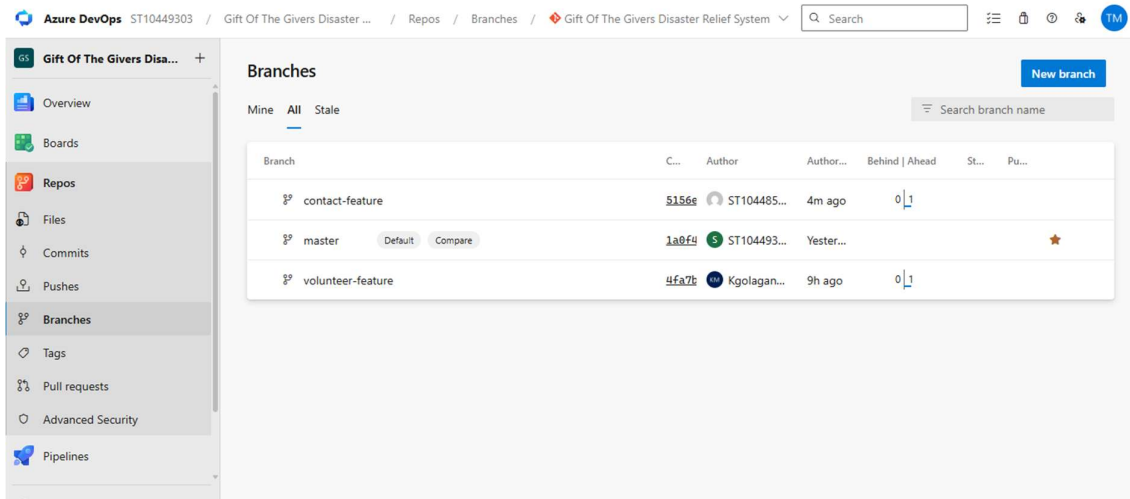


Figure 27: Visual Studio Terminal showing the home-feature branch active.

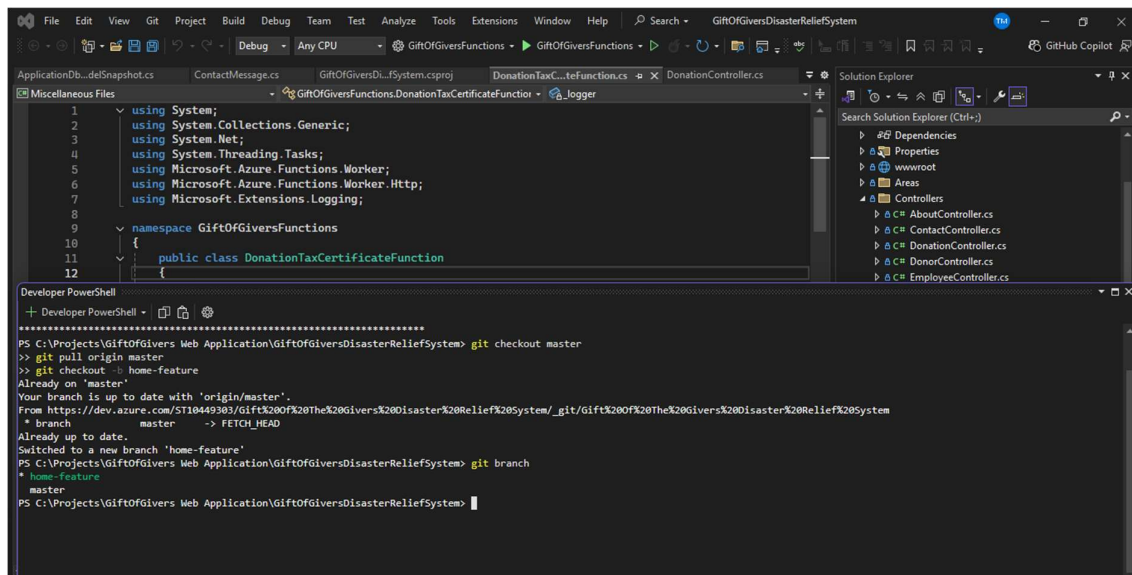


Figure 28: Home feature change detected on the home-feature branch.

The screenshot shows the Visual Studio IDE with the 'GiftOfGiversDisasterReliefSystem' project open. The 'HomeController.cs' file is selected in the Solution Explorer. The code in the file is as follows:

```
1 using System.Diagnostics;
2 using GiftOfGiversDisasterReliefSystem.Models;
3 using Microsoft.AspNetCore.Mvc;
4
5 // Home feature updated for Azure Repos branching task
6 namespace GiftOfGiversDisasterReliefSystem.Controllers
7 {
8     public class HomeController : Controller
9     {
10    }
```

The Developer PowerShell window shows the following commands and output:

```
PS C:\Projects\GiftOfGivers Web Application\GiftOfGiversDisasterReliefSystem> git branch
* branch master -> FETCH_HEAD
Already up to date.
Switched to a new branch 'home-feature'
PS C:\Projects\GiftOfGivers Web Application\GiftOfGiversDisasterReliefSystem> git add
PS C:\Projects\GiftOfGivers Web Application\GiftOfGiversDisasterReliefSystem> git commit -a
no changes added to commit (use "git add" and/or "git commit -a")
PS C:\Projects\GiftOfGivers Web Application\GiftOfGiversDisasterReliefSystem> git push
Enumerating objects: 1, done.
Counting objects: 100% (1/1), done.
Delta compression using up to 2 threads
Compressing objects: 100% (1/1), done.
Writing objects: 100% (1/1), 458 bytes | 91.00 KiB/s, done.
Total 5 (delta 4), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Analyzing objects... (5/5) (28 ms)
remote: Validating commits... (1/1) done (0 ms)
remote: Storing packfile... done (48 ms)
remote: Storing index... done (61 ms)
remote: Updating refs... done (188 ms)
remote:
remote: Create a pull request for 'home-feature' by visiting:
remote: https://dev.azure.com/ST10449303/GiftOfGiversDisasterReliefSystem/_git/GiftOfGiversDisasterReliefSystem/pullrequestcreate?sourceRef=home-feature
remote:
To https://dev.azure.com/ST10449303/GiftOfGiversDisasterReliefSystem/_git/GiftOfGiversDisasterReliefSystem
 * [new branch] home-feature -> home-feature
branch 'home-feature' set up to track 'origin/home-feature'.
PS C:\Projects\GiftOfGivers Web Application\GiftOfGiversDisasterReliefSystem>
```

Figure 29: Successful commit and push of the home-feature branch to Azure Repos.

The screenshot shows the Visual Studio IDE with the 'GiftOfGiversDisasterReliefSystem' project open. The 'HomeController.cs' file is selected in the Solution Explorer. The code in the file is as follows:

```
1 using System.Diagnostics;
2 using GiftOfGiversDisasterReliefSystem.Models;
3 using Microsoft.AspNetCore.Mvc;
4
5 // Home feature updated for Azure Repos branching task
6 namespace GiftOfGiversDisasterReliefSystem.Controllers
7 {
8     public class HomeController : Controller
9     {
10    }
```

The Developer PowerShell window shows the following commands and output:

```
PS C:\Projects\GiftOfGivers Web Application\GiftOfGiversDisasterReliefSystem> git push
Enumerating objects: 1, done.
Counting objects: 100% (1/1), done.
Delta compression using up to 2 threads
Compressing objects: 100% (1/1), done.
Writing objects: 100% (1/1), 458 bytes | 91.00 KiB/s, done.
Total 5 (delta 4), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Analyzing objects... (5/5) (28 ms)
remote: Validating commits... (1/1) done (0 ms)
remote: Storing packfile... done (48 ms)
remote: Storing index... done (61 ms)
remote: Updating refs... done (188 ms)
remote:
remote: Create a pull request for 'home-feature' by visiting:
remote: https://dev.azure.com/ST10449303/GiftOfGiversDisasterReliefSystem/_git/GiftOfGiversDisasterReliefSystem/pullrequestcreate?sourceRef=home-feature
remote:
To https://dev.azure.com/ST10449303/GiftOfGiversDisasterReliefSystem/_git/GiftOfGiversDisasterReliefSystem
 * [new branch] home-feature -> home-feature
branch 'home-feature' set up to track 'origin/home-feature'.
PS C:\Projects\GiftOfGivers Web Application\GiftOfGiversDisasterReliefSystem>
```

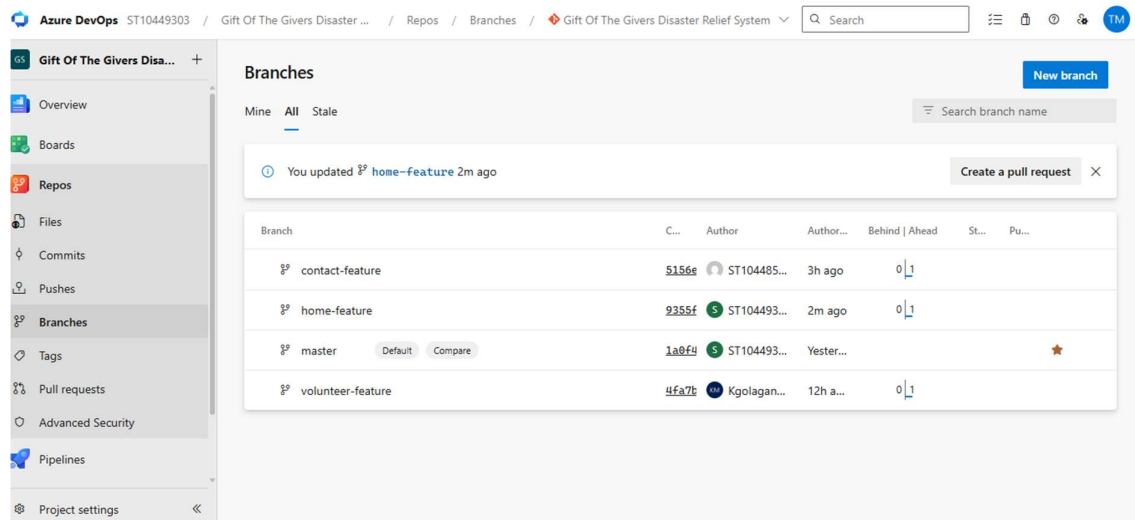


Figure 30: Visual Studio Terminal showing the about-feature branch created and active.

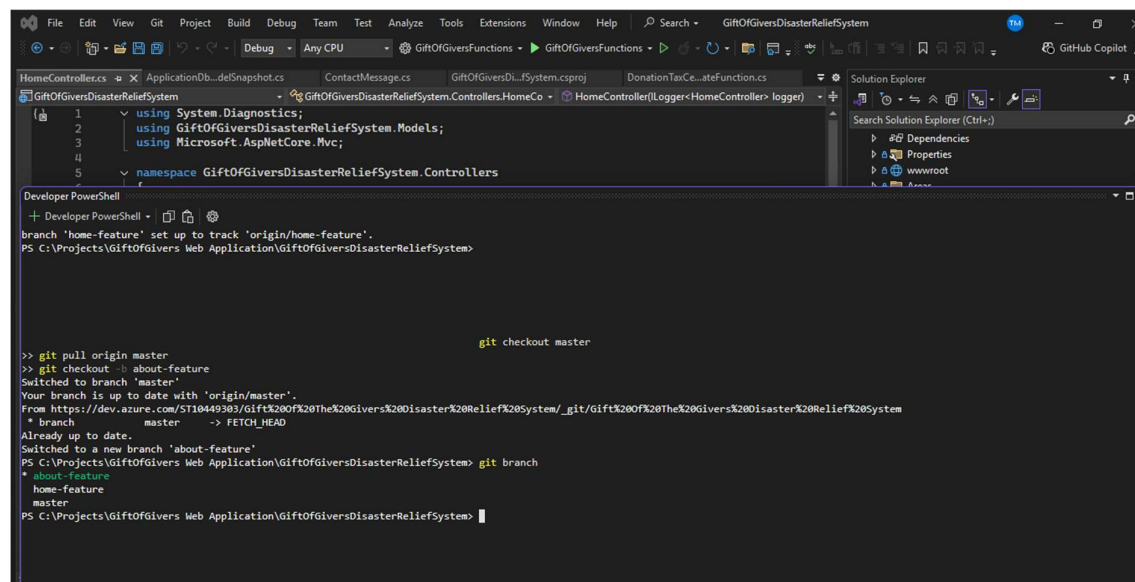
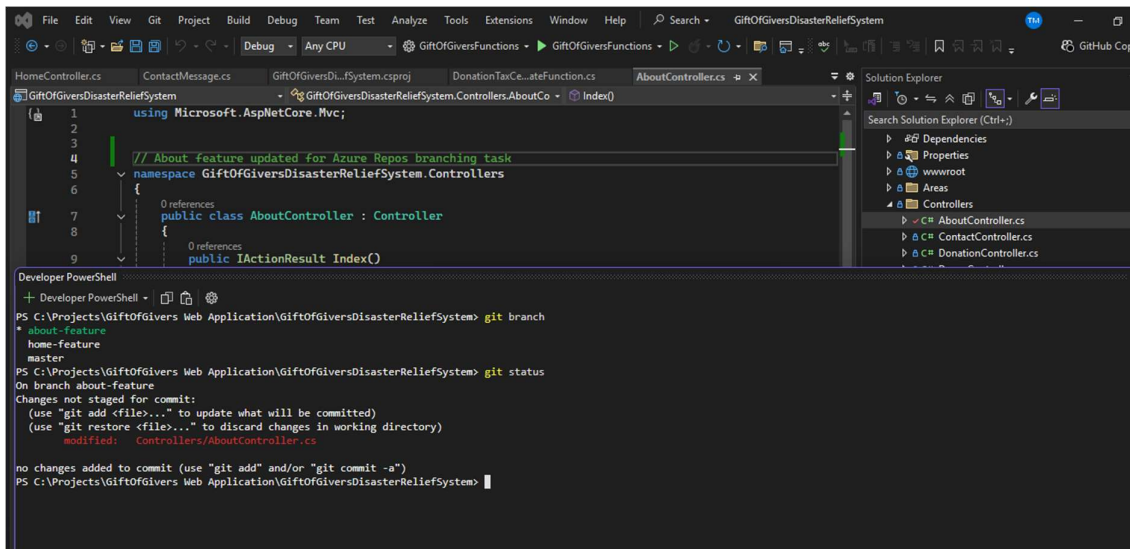


Figure 31: About feature change detected on the about-feature branch.



The screenshot shows the Visual Studio IDE with the 'GiftOfGiversDisasterReliefSystem' project open. The 'AboutController.cs' file is selected in the Solution Explorer, showing the following code:

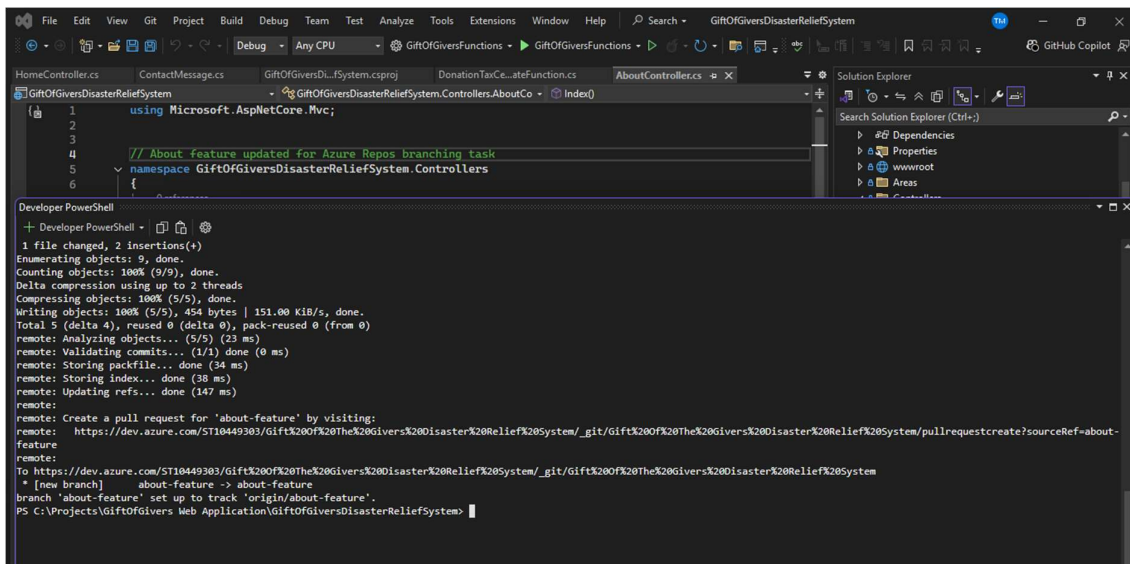
```
1 using Microsoft.AspNetCore.Mvc;
2
3
4 // About feature updated for Azure Repos branching task
5 namespace GiftOfGiversDisasterReliefSystem.Controllers
6 {
7     public class AboutController : Controller
8     {
9         public IActionResult Index()
10     }
11 }
```

The terminal window shows the following commands and output:

```
PS C:\Projects\GiftOfGivers Web Application\GiftOfGiversDisasterReliefSystem> git branch
* about-feature
home-feature
master
PS C:\Projects\GiftOfGivers Web Application\GiftOfGiversDisasterReliefSystem> git status
On branch about-feature
Changes not staged for commit:
  (use "git add <file>..." to update what will be committed)
  (use "git restore <file>..." to discard changes in working directory)
        modified:   Controllers\AboutController.cs

no changes added to commit (use "git add" and/or "git commit -a")
PS C:\Projects\GiftOfGivers Web Application\GiftOfGiversDisasterReliefSystem>
```

Figure 32: Successful commit and push of the about-feature branch to Azure Repos.



The screenshot shows the Visual Studio IDE with the 'GiftOfGiversDisasterReliefSystem' project open. The 'AboutController.cs' file is selected in the Solution Explorer, showing the following code:

```
1 using Microsoft.AspNetCore.Mvc;
2
3
4 // About feature updated for Azure Repos branching task
5 namespace GiftOfGiversDisasterReliefSystem.Controllers
6 {
7     public class AboutController : Controller
8     {
9         public IActionResult Index()
10     }
11 }
```

The terminal window shows the following commands and output:

```
1 file changed, 2 insertions(+)
Enumerating objects: 0, done.
Counting objects: 100% (9/9), done.
Delta compression using up to 2 threads
Compressing objects: 100% (5/5), done.
Writing objects: 100% (5/5), 454 bytes | 151.00 KiB/s, done.
Total 5 (delta 4), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Analyzing objects... (5/5) (23 ms)
remote: Validating commits... (1/1) done (0 ms)
remote: Storing packfile... done (34 ms)
remote: Storing index... done (38 ms)
remote: Updating refs... done (147 ms)
remote:
remote: Create a pull request for 'about-feature' by visiting:
remote:   https://dev.azure.com/ST10449303/GiftX200FX20TheX20GiversX20DisasterX20ReliefX20System/_git/GiftX200FX20TheX20GiversX20DisasterX20ReliefX20System/pullrequestcreate?sourceRef=about-
remote:
remote: To https://dev.azure.com/ST10449303/GiftX200FX20TheX20GiversX20DisasterX20ReliefX20System/_git/GiftX200FX20TheX20GiversX20DisasterX20ReliefX20System
* [new branch] about-feature -> about-feature
branch 'about-feature' set up to track 'origin/about-feature'.
PS C:\Projects\GiftOfGivers Web Application\GiftOfGiversDisasterReliefSystem>
```

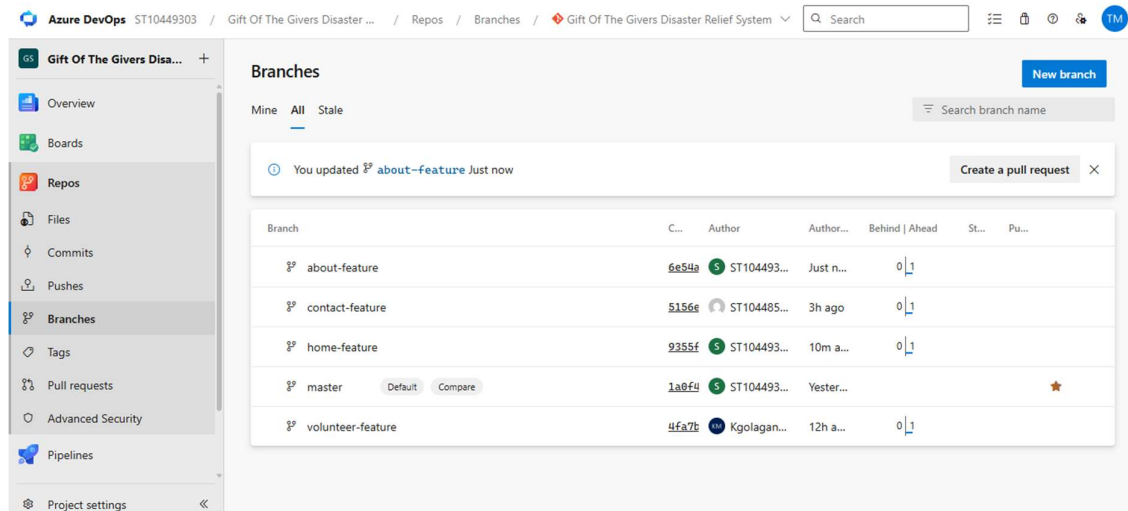


Figure 33: Visual Studio Terminal showing the employee-feature branch active.

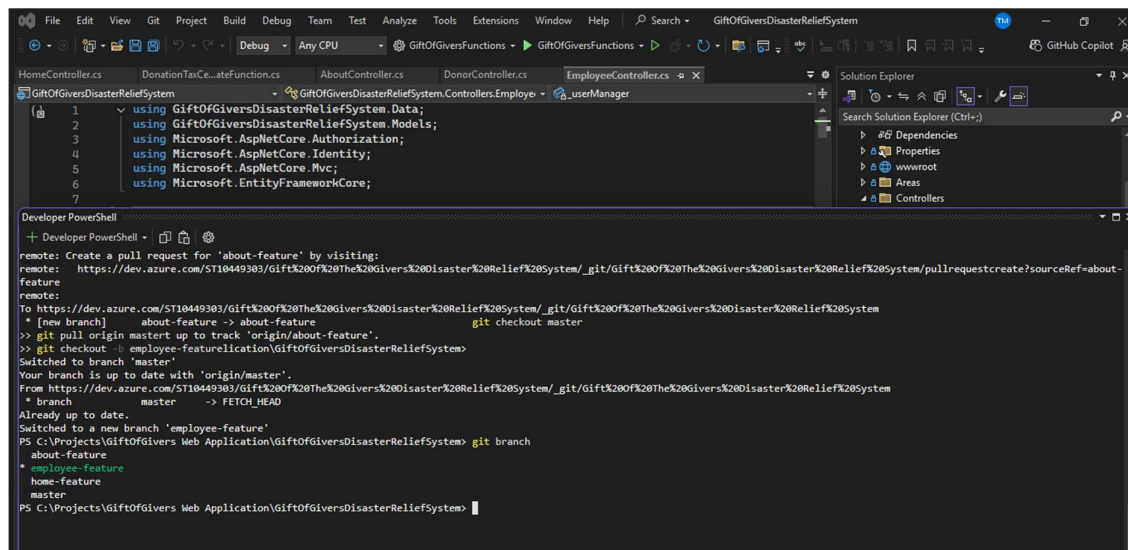


Figure 34: Employee feature change detected on the employee-feature branch.

The screenshot shows the Visual Studio IDE with the 'GiftOfGiversDisasterReliefSystem' project open. The 'EmployeeController.cs' file is selected in the Solution Explorer. The Developer PowerShell window displays the following commands and output:

```
remotes:
To https://dev.azure.com/ST10449303/Gift%20Of%20The%20Givers%20Disaster%20Relief%20System/_git/Gift%20Of%20The%20Givers%20Disaster%20Relief%20System
* [new branch] about-feature -> about-feature
>> git pull origin master up to track 'origin/about-feature'.
>> git checkout -b employee-feature/GiftOfGiversDisasterReliefSystem
Switched to branch 'master'
Your branch is up to date with 'origin/master'.
From https://dev.azure.com/ST10449303/Gift%20Of%20The%20Givers%20Disaster%20Relief%20System/_git/Gift%20Of%20The%20Givers%20Disaster%20Relief%20System
* branch master -> FETCH_HEAD
Already up to date.
Switched to a new branch 'employee-feature'
PS C:\Projects\GiftOfGivers Web Application\GiftOfGiversDisasterReliefSystem> git branch
about-feature
* employee-feature
home-feature
master
PS C:\Projects\GiftOfGivers Web Application\GiftOfGiversDisasterReliefSystem> git status
On branch employee-feature
Changes not staged for commit:
  (use "git add <file>..." to update what will be committed)
  (use "git restore <file>..." to discard changes in working directory)
    modified:   Controllers/EmployeeController.cs

no changes added to commit (use "git add" and/or "git commit -a")
PS C:\Projects\GiftOfGivers Web Application\GiftOfGiversDisasterReliefSystem>
```

Figure 35: Successful commit and push of the employee-feature branch to Azure Repos.

The screenshot shows the Visual Studio IDE with the 'GiftOfGiversDisasterReliefSystem' project open. The 'EmployeeController.cs' file is selected in the Solution Explorer. The Developer PowerShell window displays the following commands and output:

```
>> git commit -m "Update employee feature" and/or "git commit -a"
>> git push --origin employee-feature/GiftOfGiversDisasterReliefSystem
[employee-feature cdl3a24] Update employee feature
1 file changed, 1 insertion(+)
Enumerating objects: 9, done.
Counting objects: 100% (9/9), done.
Delta compression using up to 2 threads
Compressing objects: 100% (5/5), done.
Writing objects: 100% (5/5), 465 bytes | 35.00 KiB/s, done.
Total 5 (delta 4), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Analyzing objects... (5/5) (22 ms)
remote: Validating commits... (1/1) done (0 ms)
remote: Storing packfile... done (44 ms)
remote: Storing index... done (40 ms)
remote: Updating refs... done (130 ms)
remote:
remote: Create a pull request for 'employee-feature' by visiting:
remote: https://dev.azure.com/ST10449303/Gift%20Of%20The%20Givers%20Disaster%20Relief%20System/_git/Gift%20Of%20The%20Givers%20Disaster%20Relief%20System/pullrequestcreate?sourceRef=employee-feature
remote:
To https://dev.azure.com/ST10449303/Gift%20Of%20The%20Givers%20Disaster%20Relief%20System/_git/Gift%20Of%20The%20Givers%20Disaster%20Relief%20System
* [new branch] employee-feature -> employee-feature
branch 'employee-feature' set up to track 'origin/employee-feature'.
PS C:\Projects\GiftOfGivers Web Application\GiftOfGiversDisasterReliefSystem>
```

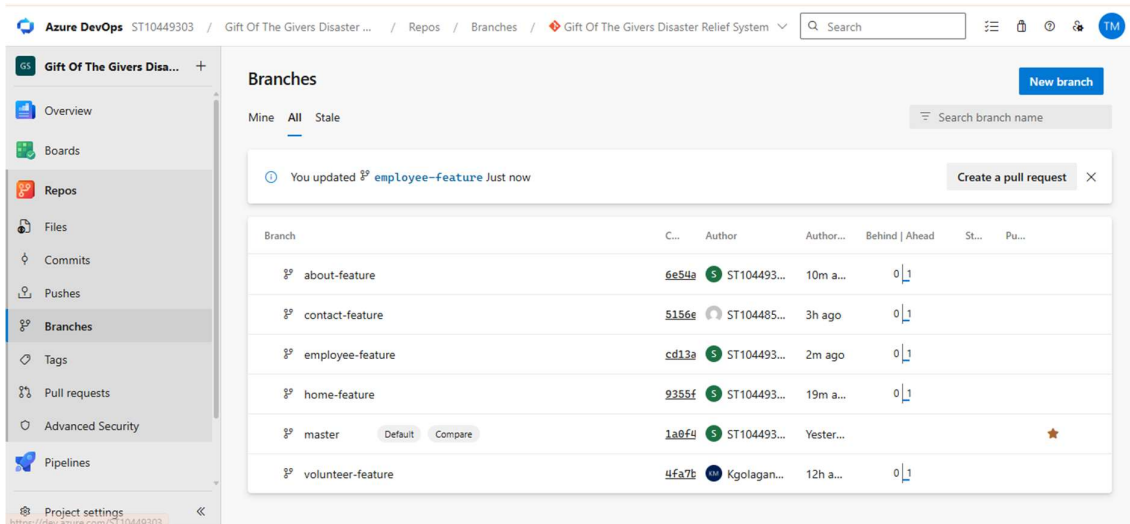


Figure 36: Successful creation and push of the main branch to Azure Repos.

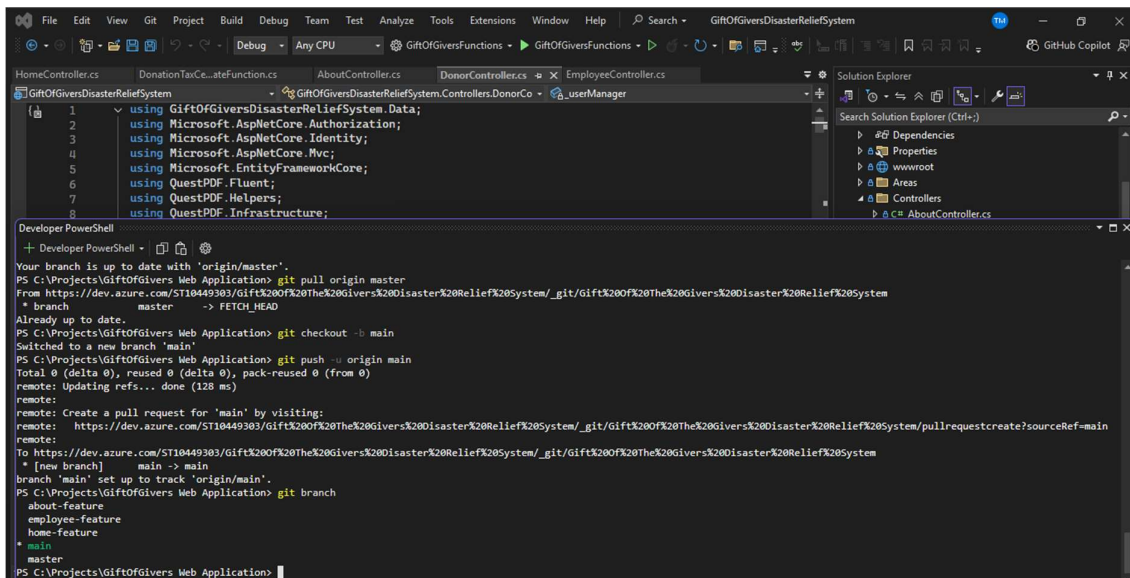


Figure 37: Visual Studio showing the merge of volunteer-feature into main.

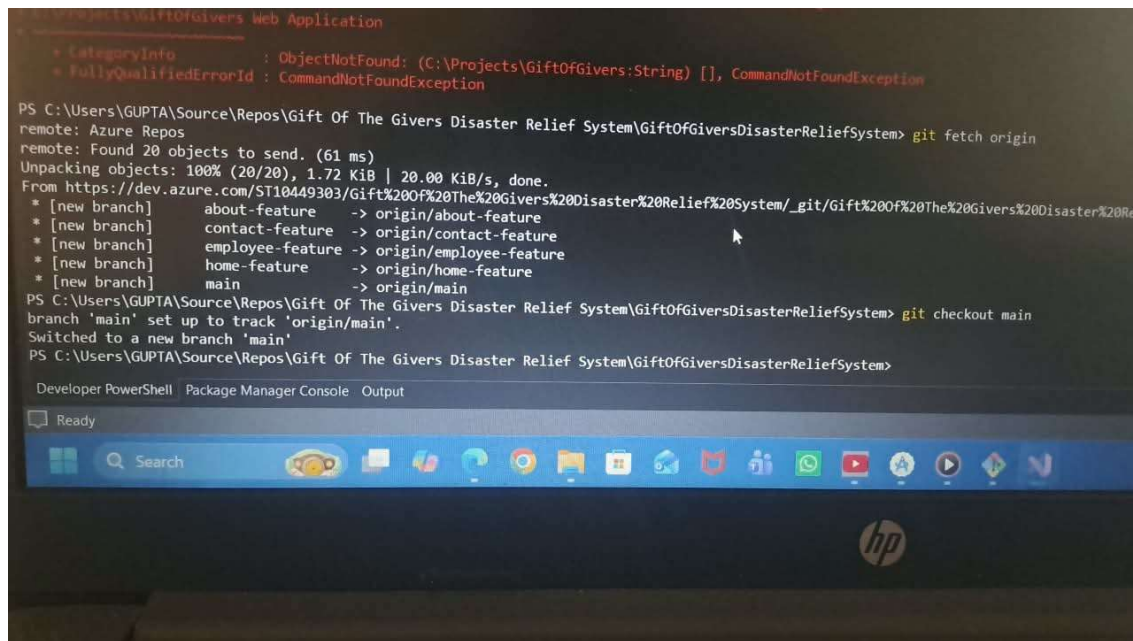


Figure 38: Visual Studio showing the volunteer-feature branch merged into the main branch.

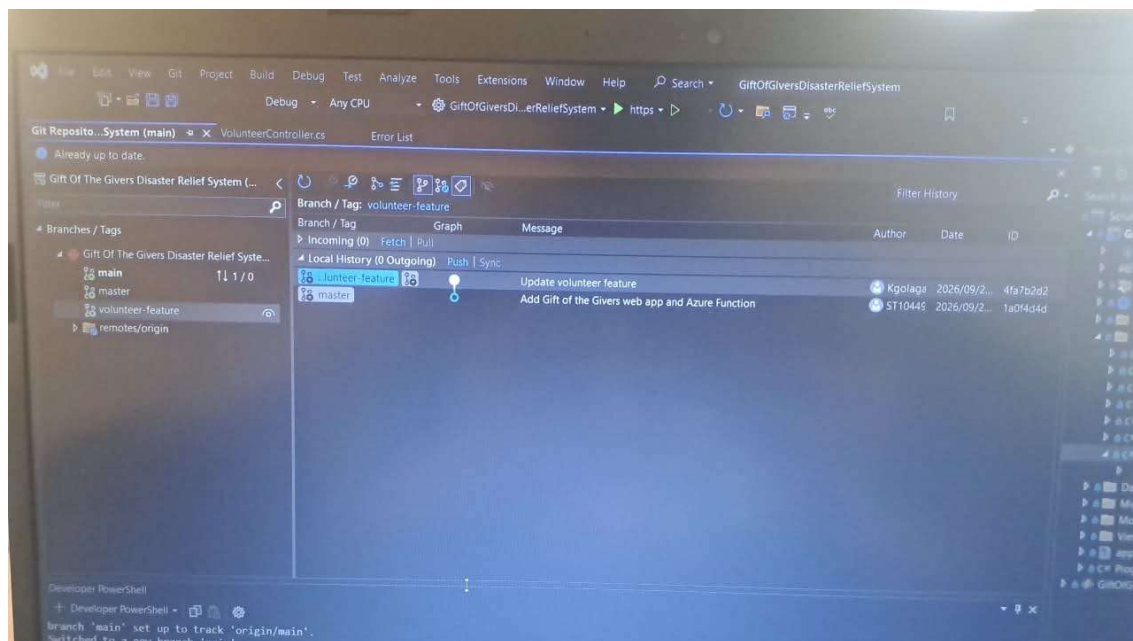


Figure 39: Successful push of the merged main branch to Azure Repos.

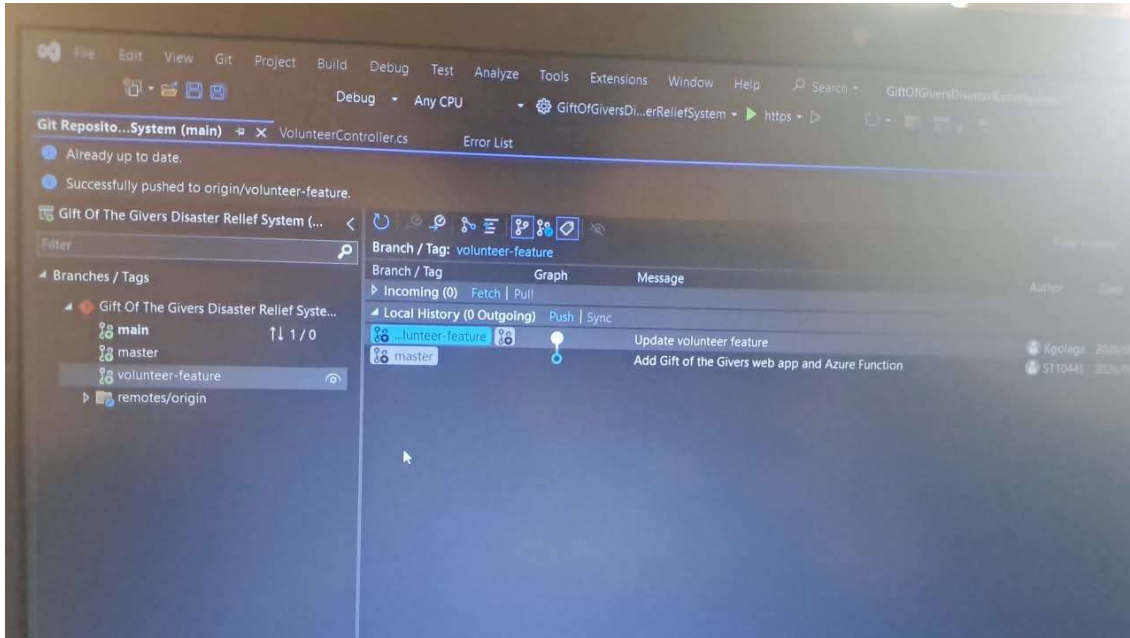
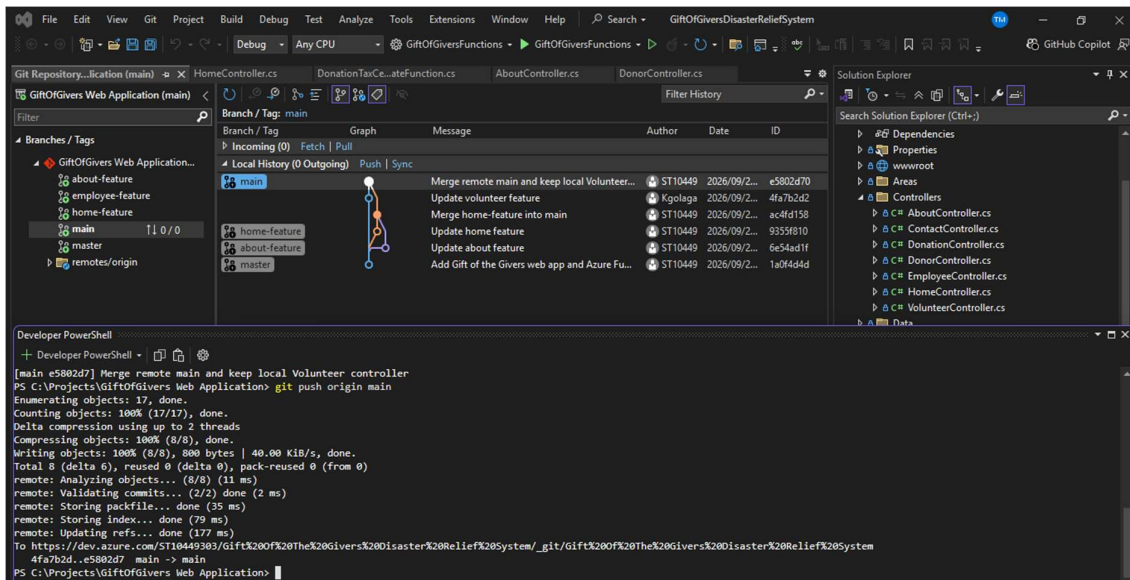


Figure 40: Azure Repos main branch showing merged feature branches and successful push to the remote repository.



Section C: Continuous Builds with Azure Pipelines

Figure 41: Azure DevOps pipeline YAML showing the Build solution and Run tests tasks.

Azure DevOps ST10449303 / Gift Of The Givers Disaster Relief System / Pipelines

Gift Of The Givers Disaster Relief System +

Overview
Boards
Repos
Pipelines
Environments
Library
Test Plans
Artifacts
Project settings

Connect Select Configure Review

New pipeline

Review your pipeline YAML

Gift Of The Givers Disaster Relief System / azure-pipelines.yml

```
23  
24 - task: DotNetCoreCLI@2  
25   displayName: 'Build solution'  
26   inputs:  
27     command: 'build'  
28     projects: '**/*.sln'  
29     arguments: '--configuration $(buildConfiguration) --no-restore'  
30  
31 - task: DotNetCoreCLI@2  
32   displayName: 'Run tests'  
33   inputs:  
34     command: 'test'  
35     projects: '**/*Tests/*.csproj'  
36     arguments: '--configuration $(buildConfiguration) --no-build'
```

Save and run

Saving will commit azure-pipelines.yml to the repository.

Commit message

Set up CI with Azure Pipelines

Optional extended description

Add an optional description...

☒ Commit directly to the master branch
☐ Create a new branch for this commit

Save and run

```
PS C:\agent> .\config.cmd  
  
Azure Pipelines  
agent v5.279.0 (commit 3898f97)  
  
>> Connect:  
Enter server URL > https://dev.azure.com/ST10449303/  
Enter authentication type (press enter for PAT) >  
Enter personal access token > *****  
Connecting to server ...  
  
>> Register Agent:  
Enter agent pool (press enter for default) >  
Enter agent name (press enter for MAHLANGUKAYISE) > GiftOfGivers-Agent  
Scanning for tool capabilities.  
Connecting to the server.  
Successfully added the agent  
Testing agent connection.  
Enter work folder (press enter for _work) >  
2026-09-24 15:24:34Z: Settings Saved.  
Enter run agent as service? (Y/N) (press enter for N) > Y  
Enter enable SERVICE_SID_TYPE_UNRESTRICTED for agent service (Y/N) (press enter for N) > N  
Error reported in diagnostic logs. Please examine the log for more details.  
- C:\agent\diag\Agent_20260924-152134-utc.log  
Needs Administrator privileges for configuring agent as windows service.  
PS C:\agent>
```

Azure DevOps ST10449303 / Gift Of The Givers Disaster Relief System / Pipelines

Gift Of The Givers Disaster Relief System +

Overview
Boards
Repos
Pipelines
Environments
Library
Test Plans
Artifacts
Project settings

← Gift Of The Givers Disaster Relief System

Variables Run

master Gift Of The Givers Disaster Relief System / azure-pipelines.yml

```
1 trigger:  
2   - main  
3  
4 pool:  
5   name: 'Default'  
6  
7 variables:  
8   buildConfiguration: 'Release'  
9  
10 steps:  
11  
12 - powershell: |  
13   Write-Host "Checking installed .NET SDK..."  
14   dotnet --info  
15   displayName: 'Check .NET SDK'  
16  
17 - powershell: |  
18   $solution = Get-Childitem -Path "$(Build.SourcesDirectory)" -Recurse -Filter "*.sln"  
19  
20   if (-not $solution) {  
21     Write-Error "No Visual Studio solution (.sln) was found."  
22     exit 1  
23   }
```

Tasks

Search tasks

- .NET Core
Build, test, package, or publish a dotnet application...
- ACR Task
Build and run container images using Azure Cont...
- Android signing
Sign and align Android APK files
- Ant
Build with Apache Ant
- App Center distribute
Distribute app builds to testers and users via Visa...
- App Center test
Test app packages with Visual Studio App Center

Azure DevOps ST10449303 / Gift Of The Givers Disaster ... / Pipelines

Gift Of The Givers Disaster Relief System

Variables Run

```
22 .....exit 1
23 .....}
24 .....}
25 .....Write-Host "Solution found: $($solution.FullName)"
26 .....dotnet restore $solution.FullName
27 .....displayName: 'Restore dependencies'
28 .....}
29 .....- powershell: |
30 .....$solution = Get-ChildItem -Path "$(Build.SourcesDirectory)" -Recurse -Filter "*"
31 .....}
32 .....if (-not $solution) {
33 .....Write-Error "No Visual Studio solution (.sln) was found."
34 .....exit 1
35 .....}
36 .....}
37 .....Write-Host "Building solution: $($solution.FullName)"
38 .....dotnet build $solution.FullName --configuration $(buildConfiguration) --no-restore
39 .....displayName: 'Build solution'
40 .....}
41 .....- powershell: |
42 .....$testProjects = Get-ChildItem -Path "$(Build.SourcesDirectory)" -Recurse -Filter "*"
43 .....}
44 .....if ($testProjects.Count -eq 0) {
45 .....Write-Host "No test projects were found in the solution."
46 .....Write-Host "Test execution step completed with no test projects configured."
47 .....exit 0
48 .....}
49 .....foreach ($testProject in $testProjects) {
50 .....Write-Host "Running tests: $($testProject.FullName)"
51 .....dotnet test $testProject.FullName --configuration $(buildConfiguration) --no-restore
52 .....if ($LASTEXITCODE -ne 0) {
53 .....exit $LASTEXITCODE
54 .....}
55 .....}
56 .....}
57 .....displayName: 'Run tests'
```

Tasks

- .NET Core: Build, test, package, or publish a dotnet application...
- ACR Task: Build and run container images using Azure Container Registry...
- Android signing: Sign and align Android APK files
- Ant: Build with Apache Ant
- App Center distribute: Distribute app builds to testers and users via Visual Studio App Center
- App Center test: Test app packages with Visual Studio App Center

#20260924.2 • Update azure-pipelines.yml for Azure Pipelines

Gift Of The Givers Disaster Relief System

Summary

Manually run by **Thabelo Mahada**

Just now

Repository and version: Gift Of The Givers Disaster Relief System, master dbf190fa

Related: 0 work items, 0 published

Tests and coverage: Get started

This pipeline needs permission to access a resource before this run can continue

Stages Jobs

Stage: Waiting

Figure 42: Azure DevOps pipeline run after granting permission to the agent pool.

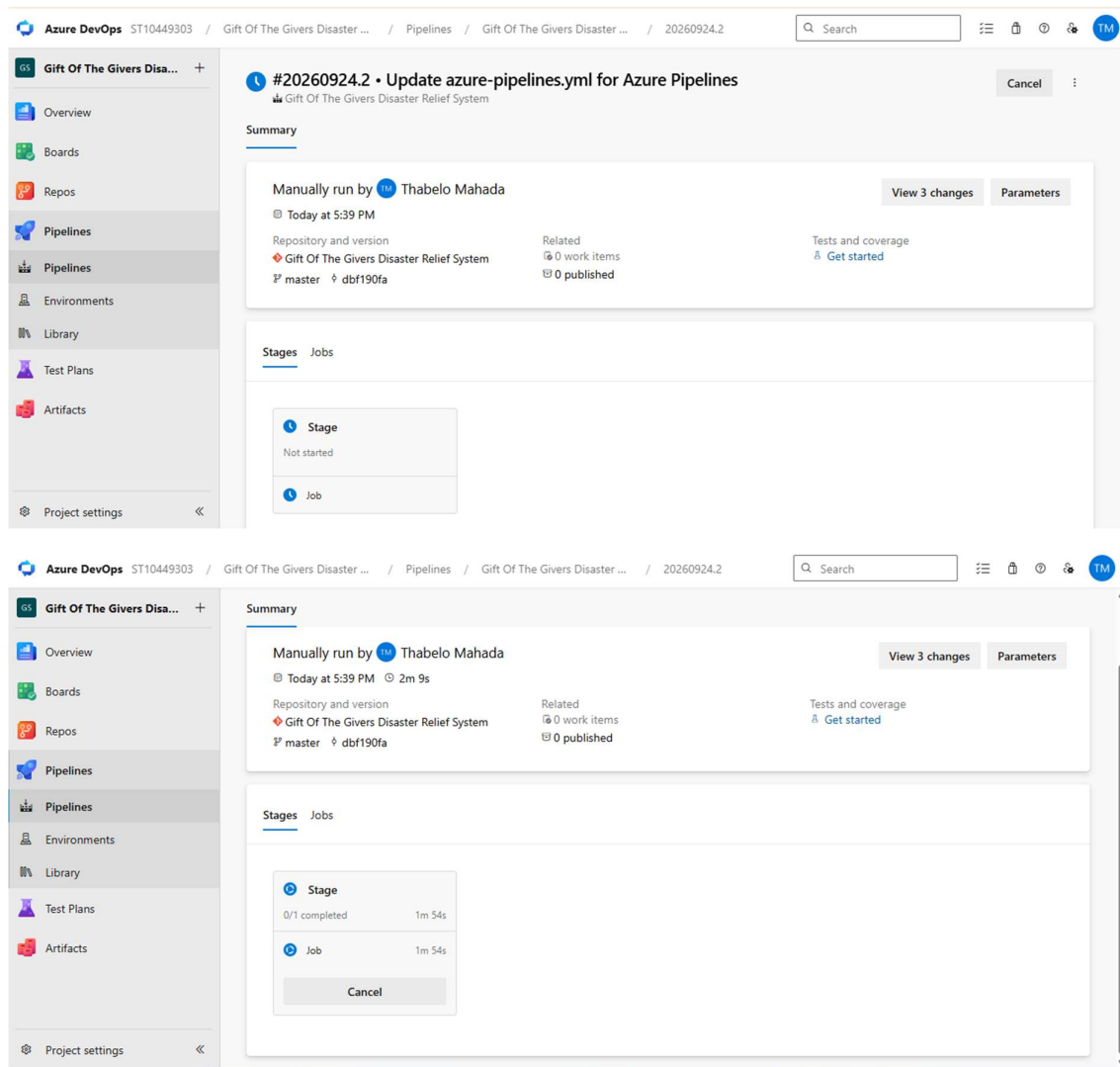


Figure 43: Azure DevOps pipeline successfully completing the build and test job on the self-hosted GiftOfGivers-Agent.

Azure DevOps ST10449303 / Gift Of The Givers Disaster ... / Pipelines / Gift Of The Givers Disaster ... / 20260924.5

Gift Of The Givers Disa... +

- Overview
- Boards
- Repos
- Pipelines
- Pipelines
- Environments
- Library
- Test Plans
- Artifacts

Jobs in run #20260924.5

Gift Of The Givers Disaster Relief System

Jobs

Job	Duration
Initialize job	4s
Checkout Gift Of The ...	50s
Check .NET SDK	55s
Restore dependenci...	1m 5s
Build solution	4m 53s
Run tests	36s
Post-job: Checkout Gift ...	4s
Finalize Job	<1s
Report build status	<1s

Job

View raw log

```
1 Pool: Default
2 Agent: GiftOfGivers-Agent
3 Started: Today at 6:24 PM
4 Duration: 8m 41s
5
6 ▶ Condition evaluation
11
12 ▶ Job preparation parameters
47 Job live console data:
48 Finishing: Job
```

Azure DevOps ST10449303 / Gift Of The Givers Disaster ... / Pipelines / Gift Of The Givers Disaster ... / 20260924.5

Gift Of The Givers Disa... +

- Overview
- Boards
- Repos
- Pipelines
- Pipelines
- Environments
- Library
- Test Plans
- Artifacts

Summary

Free memory is lower than 5% Currently used: 95.32%
Build solution

Free memory is lower than 5% Currently used: 95.32%
Build solution

Free memory is lower than 5% Currently used: 95.32%
Build solution

Stages Jobs

Stage	Jobs	Duration
Stage	1 job completed	8m 55s
Job		8m 41s

Rerun Stage

dev.azure.com/ST10449303/Gift%20Of%20The%20Givers%20Disaster%20Relief%20System/_build?definitionId=18_a=summary

Azure DevOps ST10449303 / Gift Of The Givers Disaster ... / Pipelines / Gift Of The Givers Disaster ...

Gift Of The Givers Disa... +

- Overview
- Boards
- Repos
- Pipelines
- Pipelines
- Environments
- Library
- Test Plans
- Artifacts

Gift Of The Givers Disaster Relief System

Edit Run pipeline

Runs Stages Branches Analytics

Description	Stages	Time
#20260924.5 • Update azure-pipelines.yml for Azure Pipelin... Manually run by master 84084d80	✓	16m ago 9m 13s
#20260924.4 • Update azure-pipelines.yml for Azure Pipelin... Manually run by master 84084d80	⌚	33m ago 9m 45s
#20260924.3 • Update azure-pipelines.yml for Azure Pipelin... Manually run by master 84084d80	✗	39m ago 1m 5s
#20260924.2 • Update azure-pipelines.yml for Azure Pipelin... Manually run by master dbf190fa	⌚	1h ago 12m 14s
#20260924.1 • Set up CI with Azure Pipelines Individual CI for master 618824a9	✗	2h ago <1s

C.2 Trigger a Build from Visual Studio Commit

Figure 44: Visual Studio Git Changes showing the C.2 code change committed with the message “Test automatic Azure Pipeline trigger”.

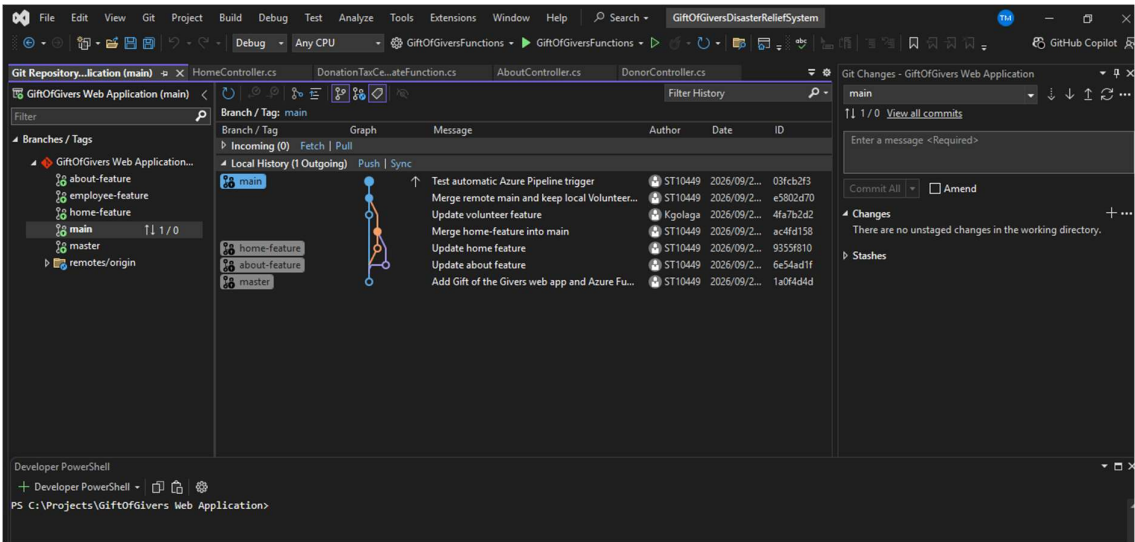
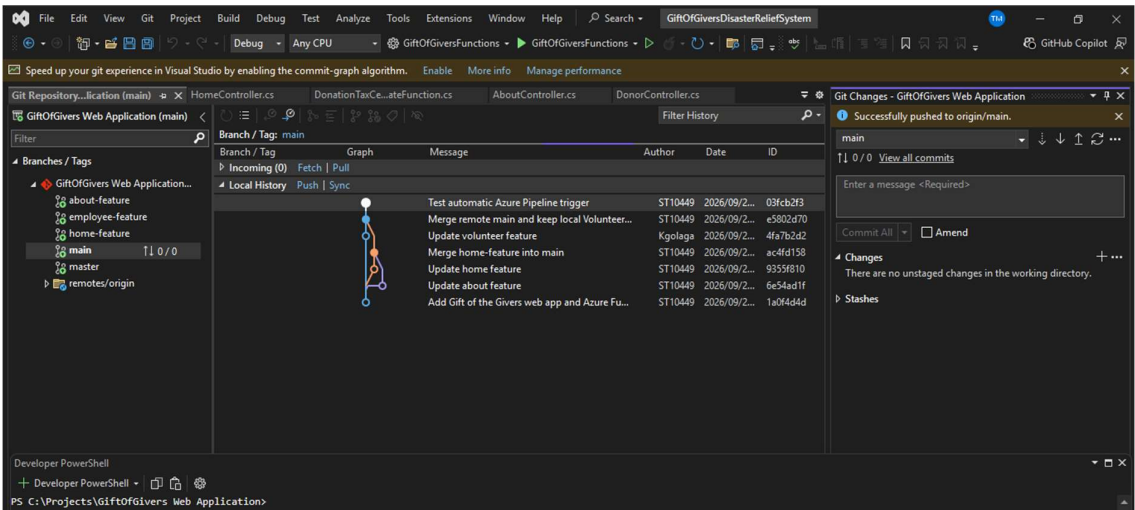


Figure 45: Visual Studio showing the C.2 commit being successfully pushed to Azure Repos.



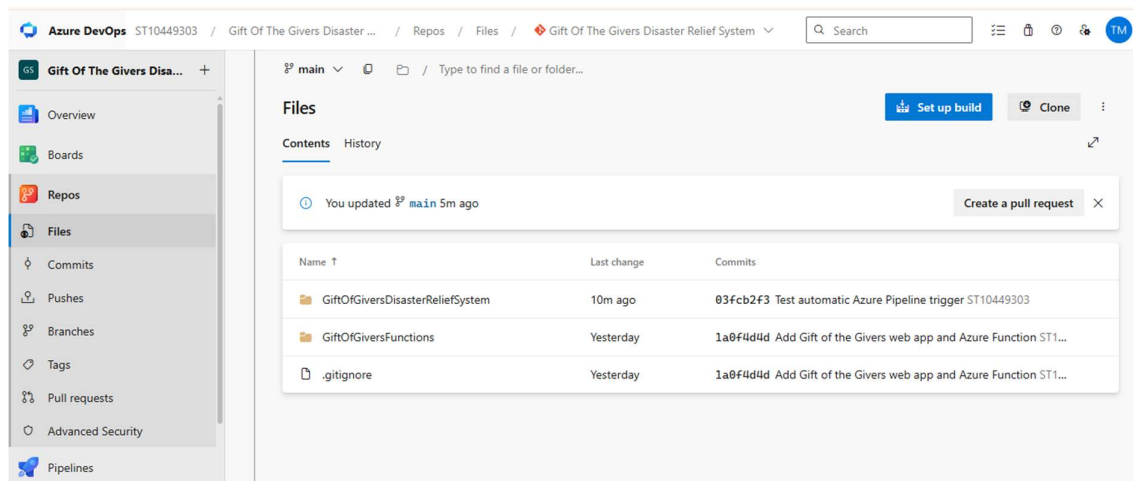


Figure 46: Solution folder showing the azure-pipelines.yml file alongside the Gift of the Givers web application and Azure Functions projects.

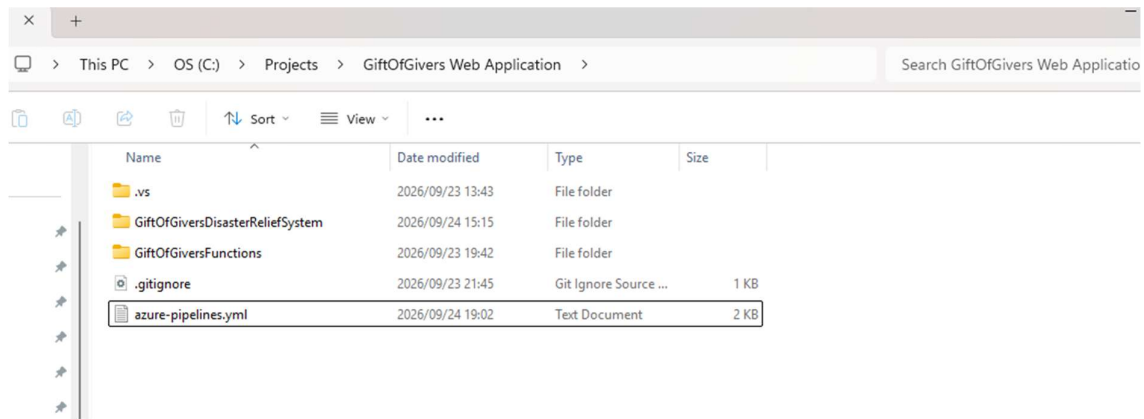
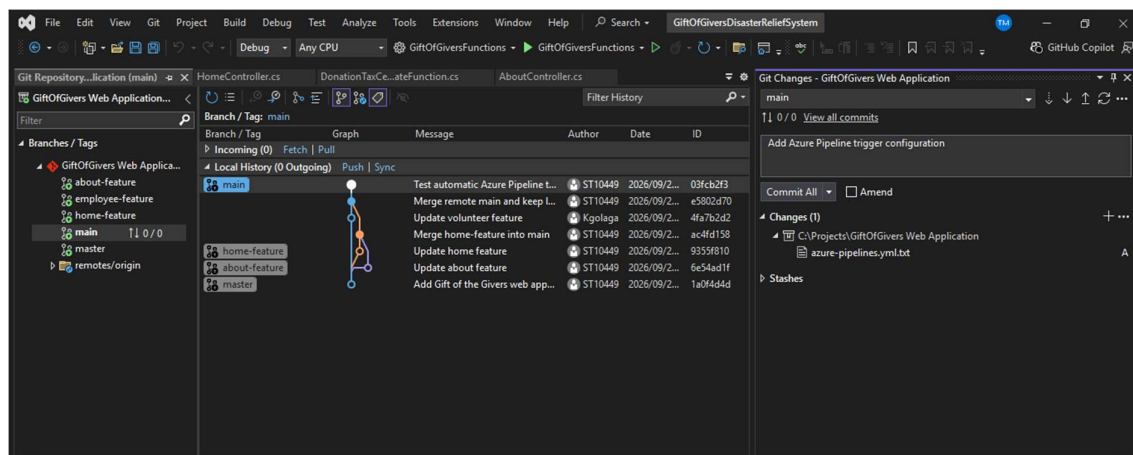


Figure 47: Visual Studio Git Changes showing the azure-pipelines.yml pipeline configuration committed to the main branch.



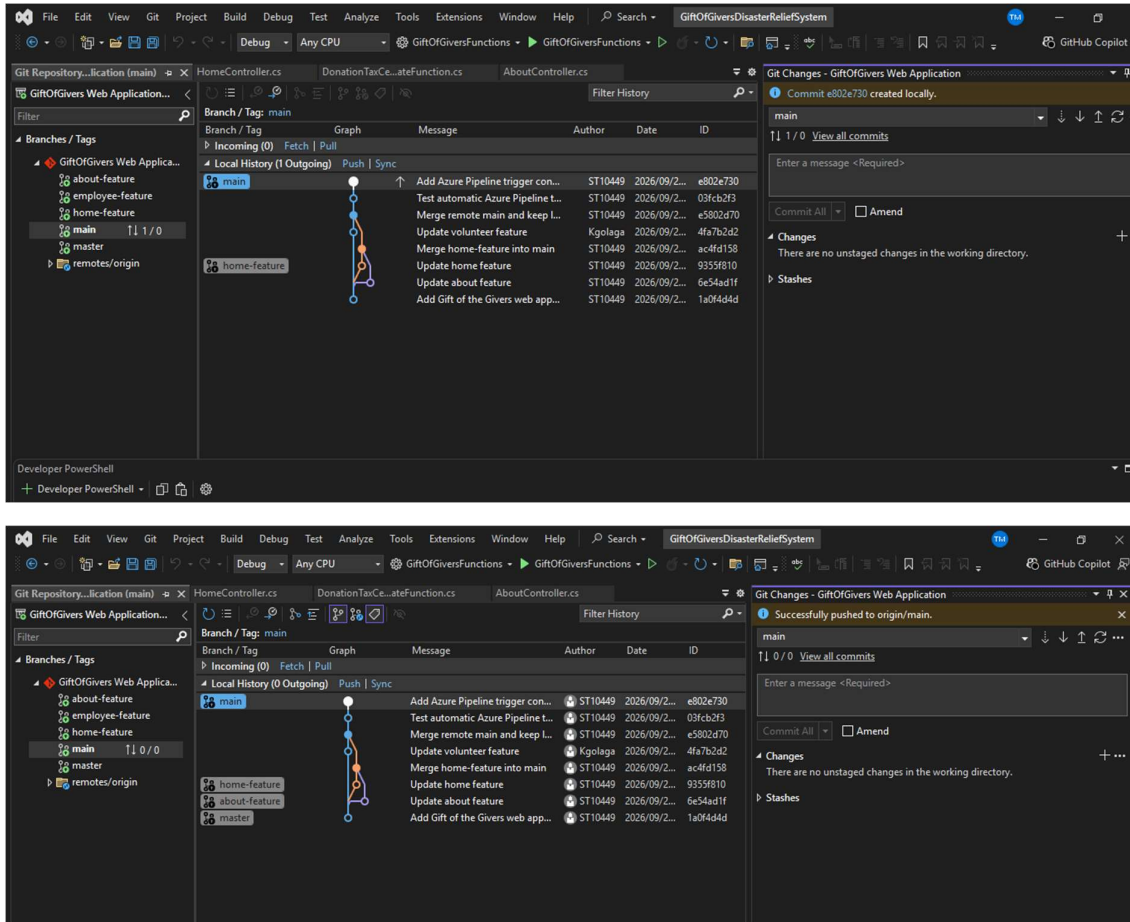


Figure 48: Azure Repos main branch showing the correctly named azure-pipelines.yml file and the latest fix commit.

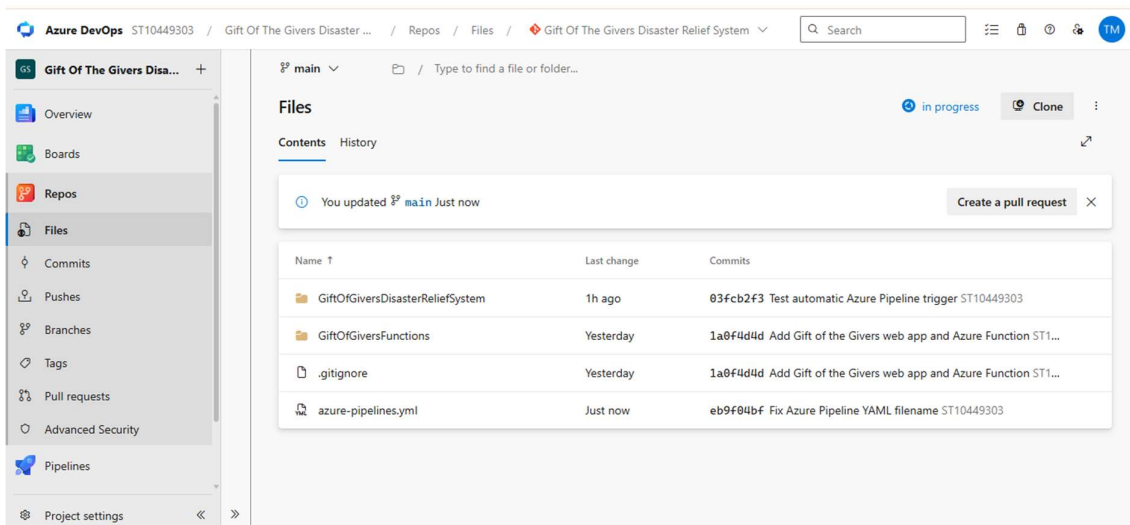


Figure 49: Azure DevOps showing the pipeline automatically triggered by the main branch commit.

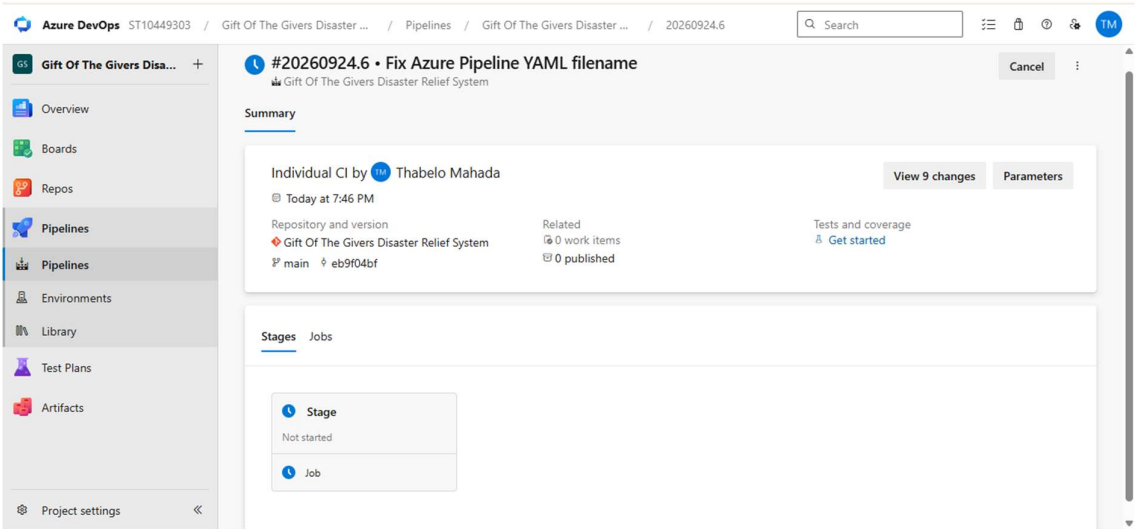
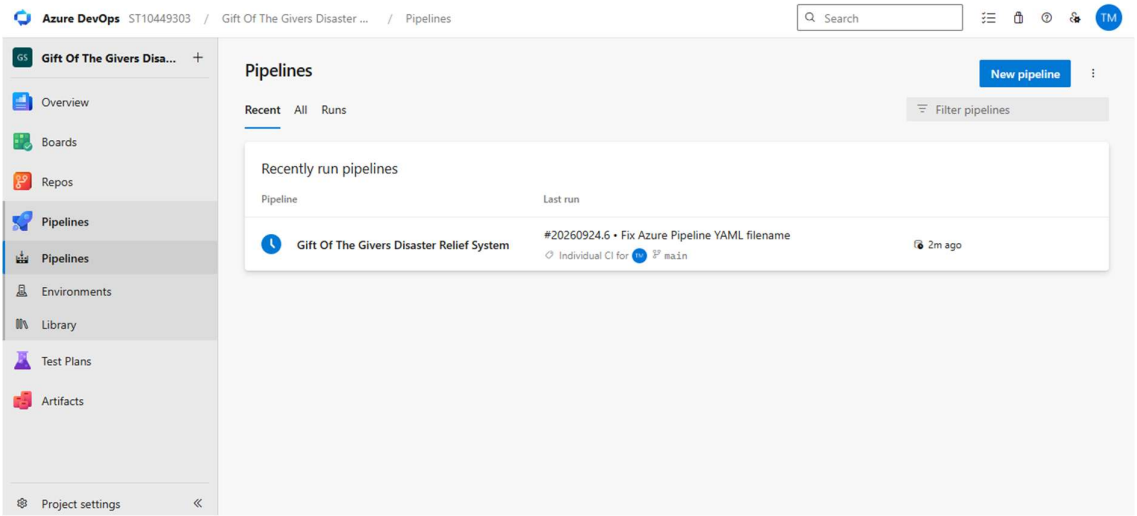


Figure 50: Azure DevOps showing the automatically triggered C.2 pipeline run on the main branch.

Azure DevOps ST10449303 / Gift Of The Givers Disaster ... / Pipelines / Gift Of The Givers Disaster ... / 20260924.6

Gift Of The Givers Disa... +

- Overview
- Boards
- Repos
- Pipelines
- Pipelines
- Environments
- Library
- Test Plans
- Artifacts

Project settings <<

Jobs in run #20260924.6

Gift Of The Givers Disaster Relief System

Jobs

Job	Duration
Initialize job	1s
Checkout Gift Of The ...	46s
Check .NET SDK	28s
Restore dependencies	37s
Build solution	2m 53s
Run tests	53s
Post-job: Checkout Gift O...	

Run tests

View task inputs

```
1 > Condition evaluation
9 Starting: Run tests
10 =====
11 Task : PowerShell
12 Description : Run a PowerShell script on Linux, macOS, or Windows
13 Version : 2.279.1
14 Author : Microsoft Corporation
15 Help : https://docs.microsoft.com/azure/devops/pipelines/tasks/utility/powershell
16 =====
17 Generating script.
18 ===== Starting Command Output =====
19 ##[warning]Free memory is lower than 5%; Currently used: 95.22%
20 "C:\WINDOWS\System32\WindowsPowerShell\v1.0\powershell.exe" -NoLogo -NoProfile -NonInteractive -Execu
```

Figure 51: Azure DevOps showing the automatically triggered pipeline successfully completing the build and test job on the main branch.

Azure DevOps ST10449303 / Gift Of The Givers Disaster ... / Pipelines / Gift Of The Givers Disaster ... / 20260924.6

Gift Of The Givers Disa... +

- Overview
- Boards
- Repos
- Pipelines
- Pipelines
- Environments
- Library
- Test Plans
- Artifacts

Project settings <<

Jobs in run #20260924.6

Gift Of The Givers Disaster Relief System

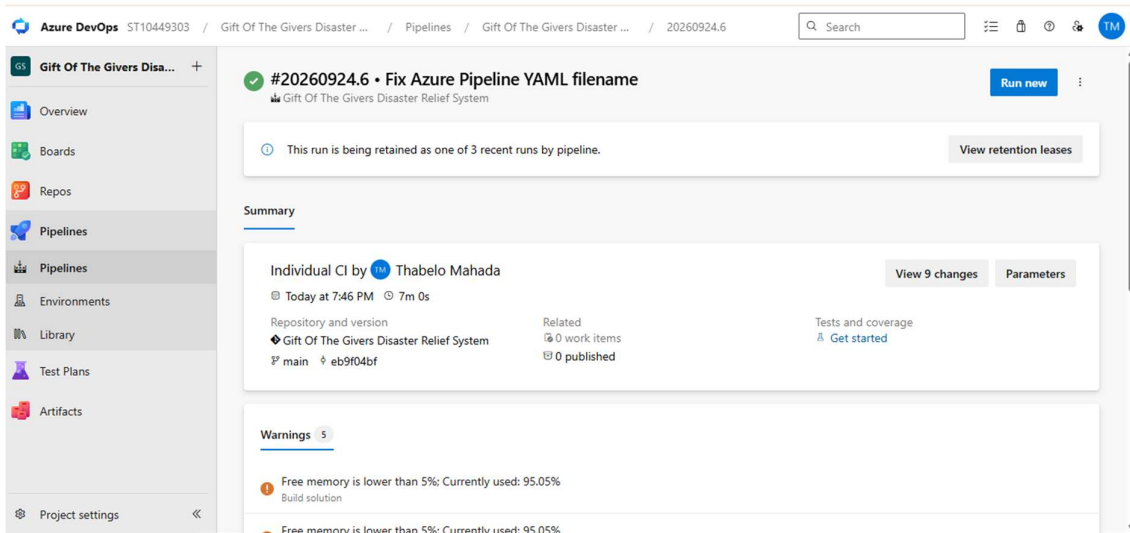
Jobs

Job	Duration
Initialize job	1s
Checkout Gift Of The ...	46s
Check .NET SDK	28s
Restore dependencies	37s
Build solution	2m 53s
Run tests	1m 28s
Post-job: Checkout Gift ...	3s
Finalize Job	<1s
Report build status	<1s

Job

View raw log

```
1 Pool: Default
2 Agent: GiftOfGivers-Agent
3 Started: Today at 7:54 PM
4 Duration: 6m 24s
5
6 > Condition evaluation
11
12 > Job preparation parameters
47 Job live console data:
48 Finishing: Job
```



Section D: Package Hosting with Azure Artifacts

D.1 Create a Custom Helper Library in Visual Studio

Figure 52: Visual Studio configuration of the GiftOfTheGivers.Helpers Class Library project.

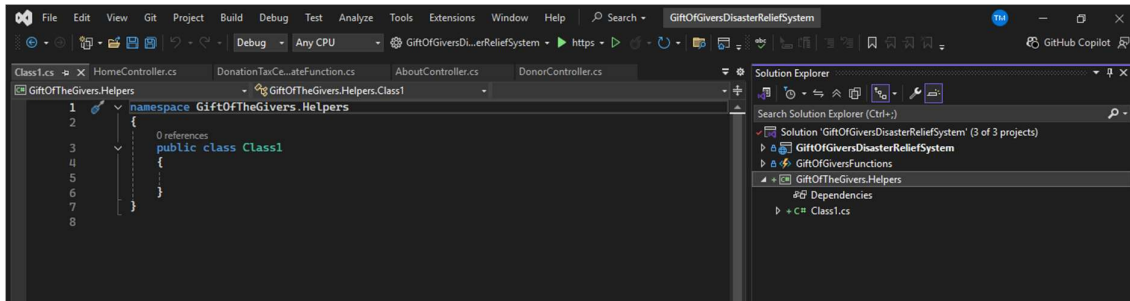


Figure 53: DonationHelper utility class containing tax certificate formatting and donation total calculation methods.

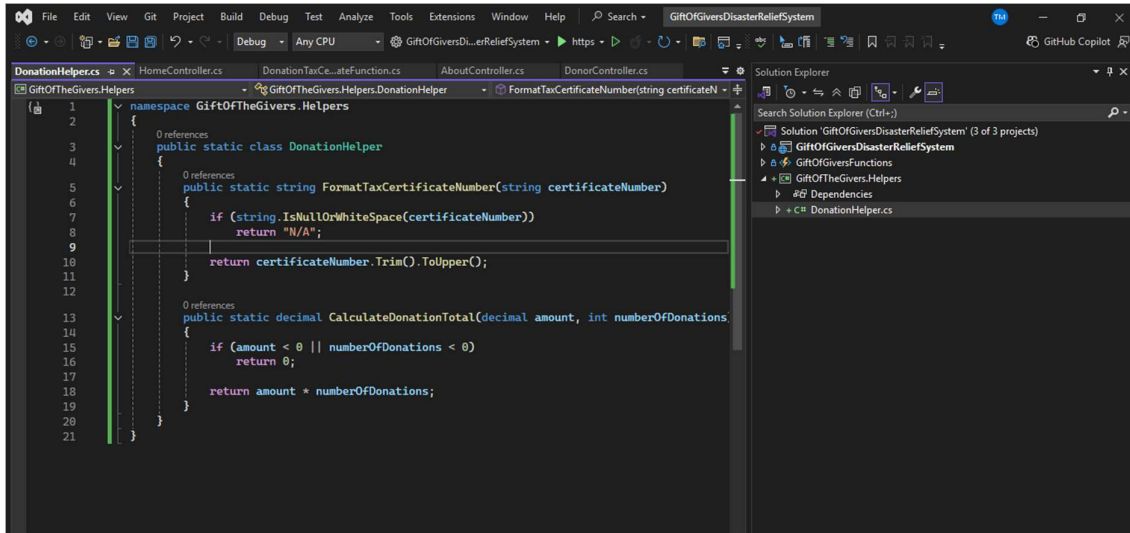


Figure 54: Successful build of the GiftOfTheGivers.Helpers Class Library.

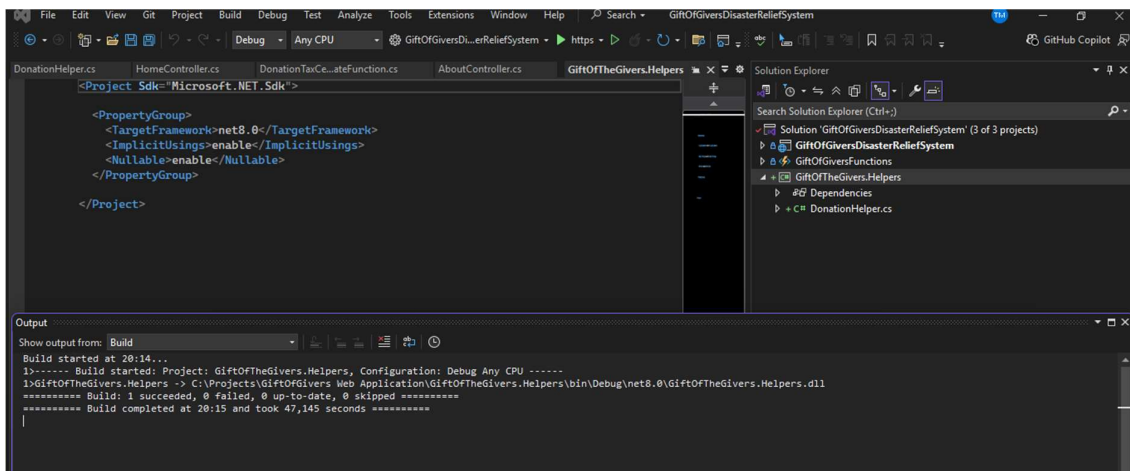


Figure 55: Successful build of the GiftOfTheGivers.Helpers project and generation of the GiftOfTheGivers.Helpers.1.0.0.nupkg package.

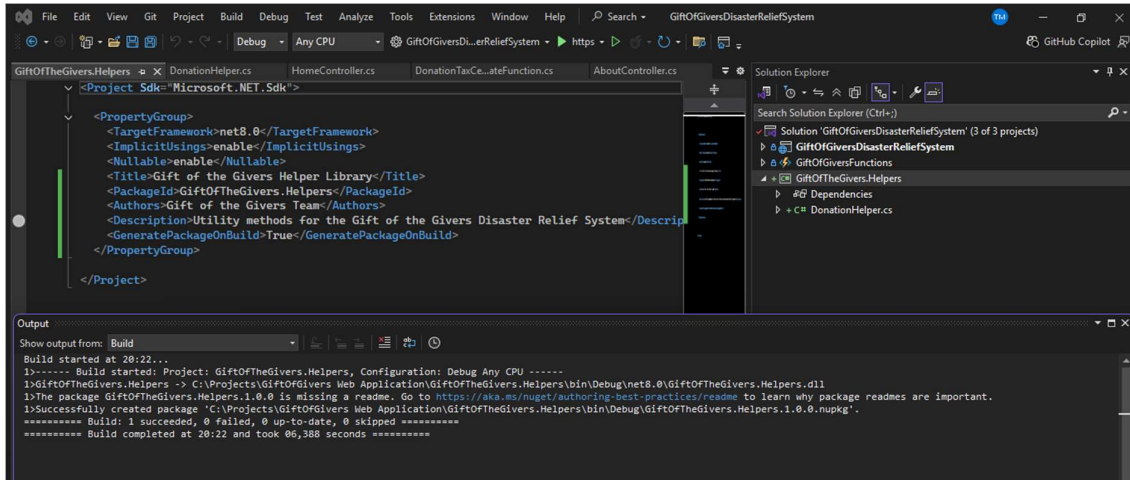


Figure 56: Azure Artifacts feed configuration for the Gift of the Givers Disaster Relief System.

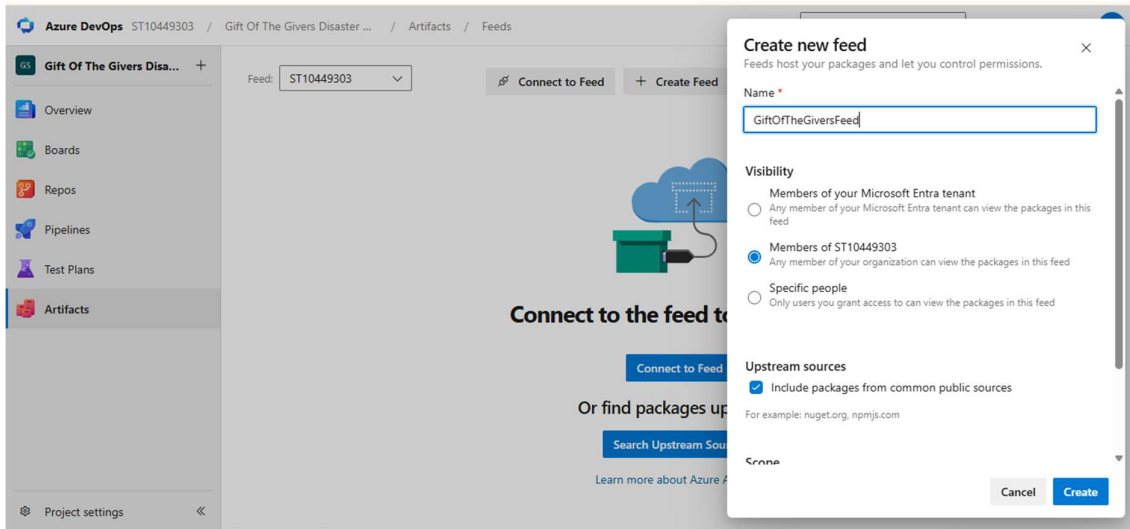


Figure 57: Azure Artifacts feed created for the Gift of the Givers Disaster Relief System.

Azure DevOps ST10449303 / Gift Of The Givers Disaster ... / Artifacts / Feeds

Search

Gift Of The Givers Disa... +

Overview
Boards
Repos
Pipelines
Test Plans
Artifacts

Feed: GiftOfTheGiversF... ▾

Connect to Feed + Create Feed Search Upstream Sources Recycle Bin



Connect to the feed to get started

Connect to Feed

Or find packages upstream

Search Upstream Sources

Create new feed



Feeds host your packages and let you control permissions.

- ☐ Any member of your Microsoft Entra tenant can view the packages in this feed
- ☒ **Members of ST10449303**
Any member of your organization can view the packages in this feed
- ☐ **Specific people**
Only users you grant access to can view the packages in this feed

Upstream sources

- ☒ Include packages from common public sources

For example: nuget.org, npmjs.com

Scope

- Project: Gift Of The Givers Disaster Relief System
(Recommended)
- ☒ The feed will be scoped to the Gift Of The Givers Disaster Relief System project.
- ☐ Organization

Cancel

Create

Figure 58: Visual Studio NuGet Package Sources configured with the Azure Artifacts GiftOfTheGiversFeed.

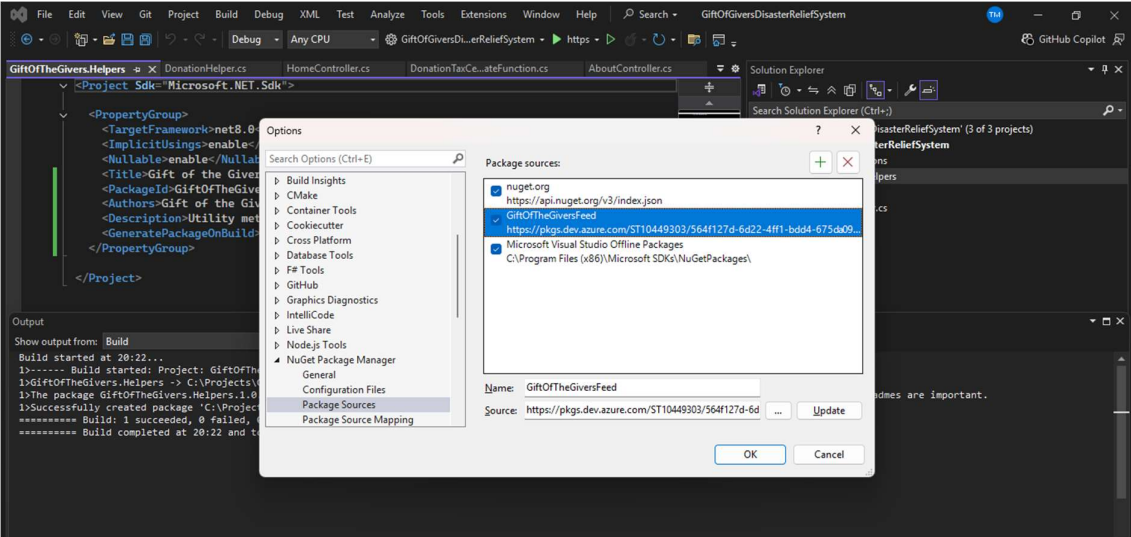


Figure 59: Successful publication of the GiftOfTheGivers.Helpers.1.0.0 NuGet package to the Azure Artifacts GiftOfTheGiversFeed.

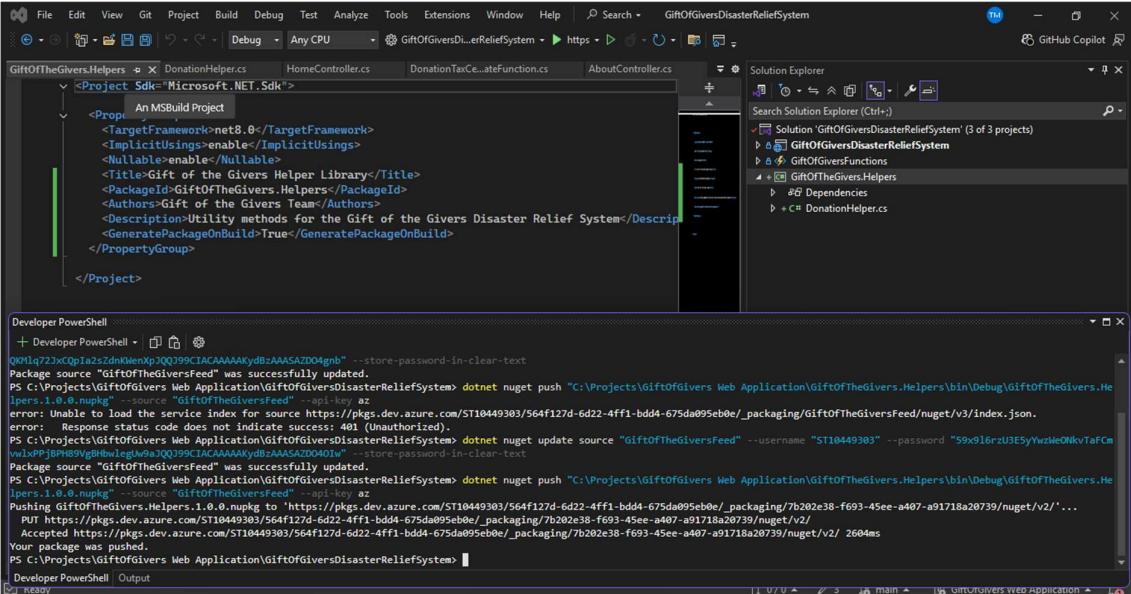
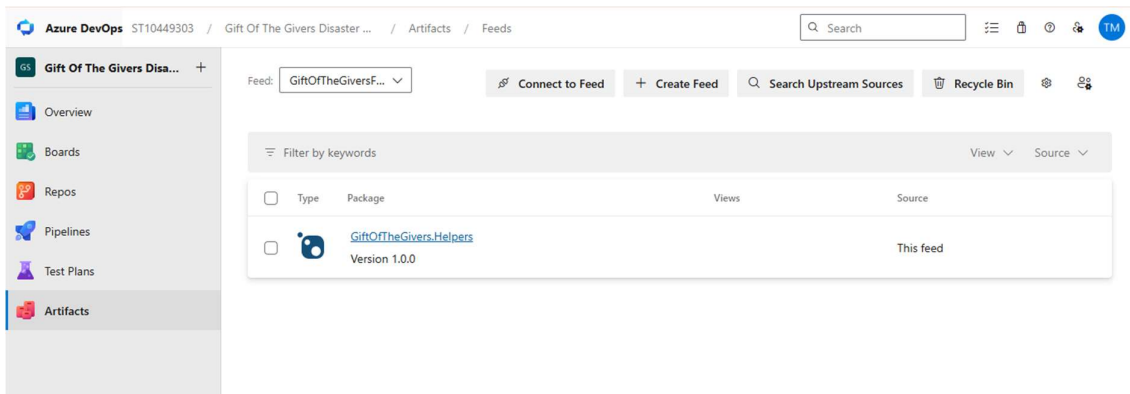


Figure 60: Azure Artifacts feed showing the published GiftOfTheGivers.Helpers NuGet package, version 1.0.0.



D.2 Use the NuGet Package in Your Web App

Figure 61: ASP.NET Core web application referencing the GiftOfTheGivers.Helpers NuGet package version 1.0.0.

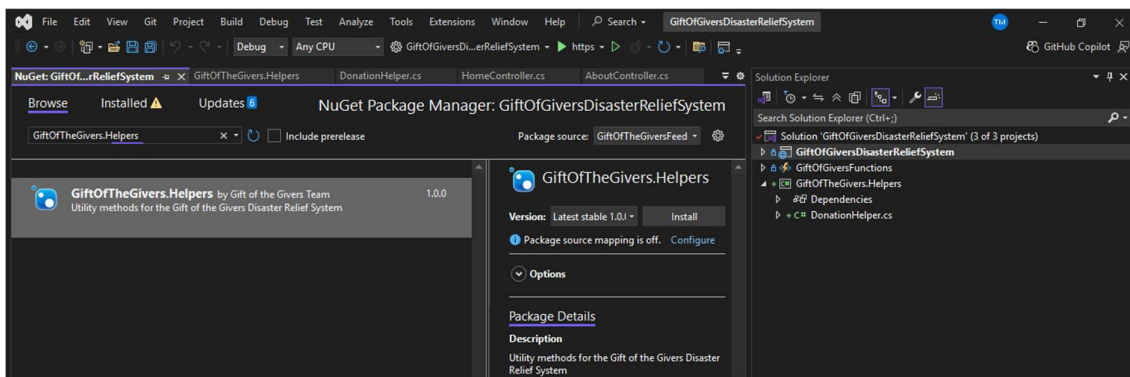


Figure 62: ASP.NET Core web application with GiftOfTheGivers.Helpers version 1.0.0 installed from the Azure Artifacts feed.

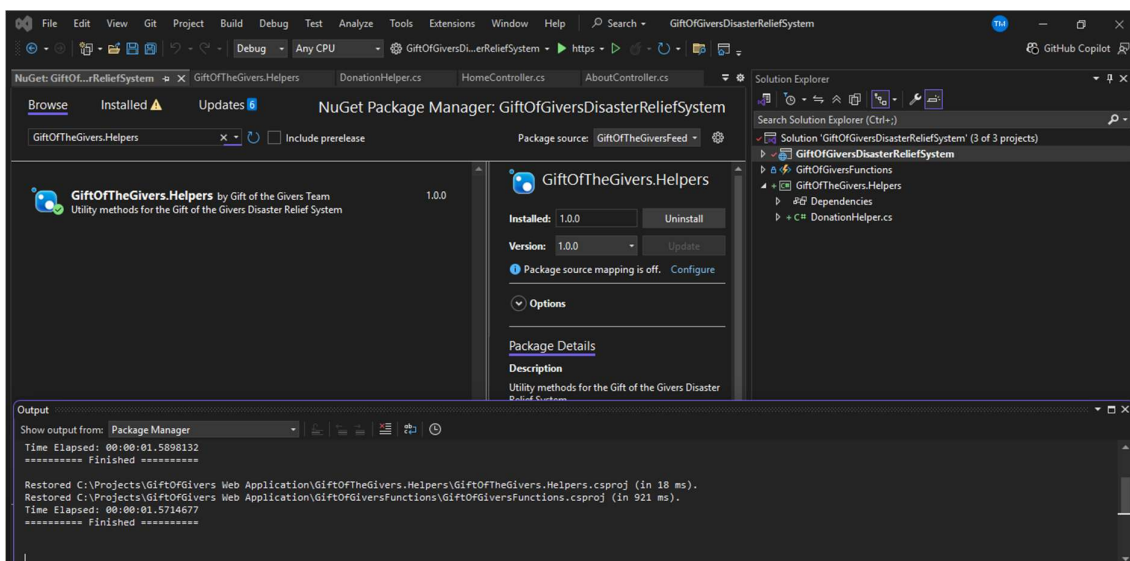


Figure 63: Successful build of the Gift of the Givers web application and Azure Functions after integrating the GiftOfTheGivers.Helpers NuGet package.

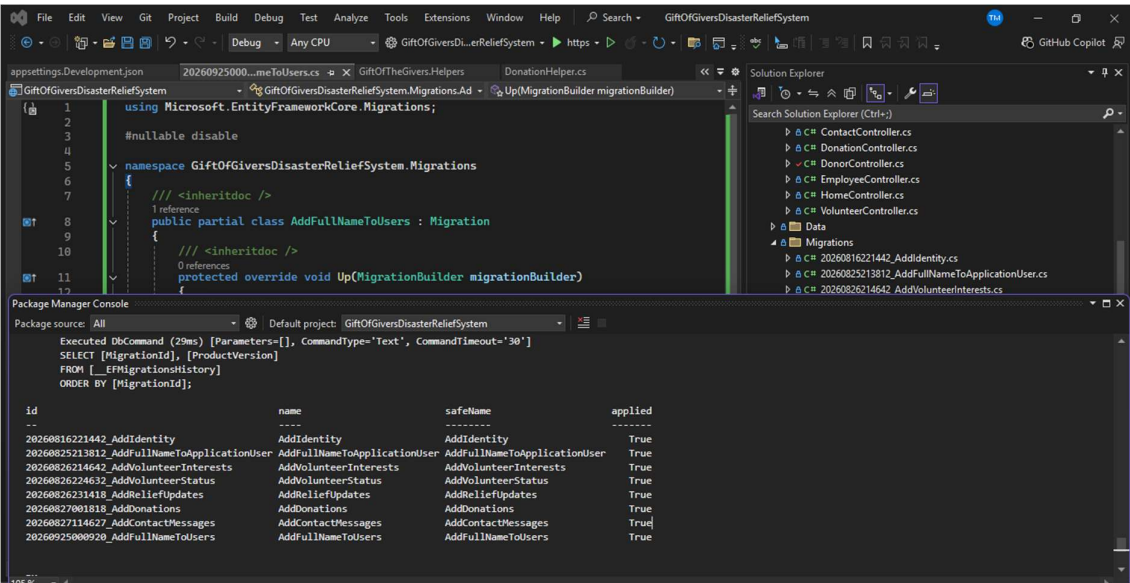
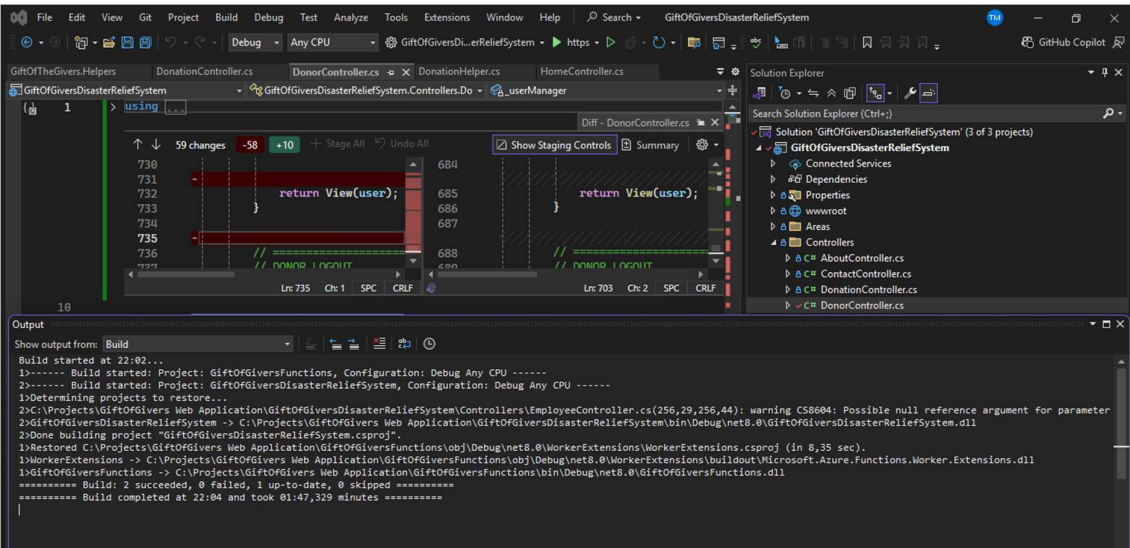


Figure 64: Successful donor donation and tax certificate generated using the integrated GiftOfTheGivers.Helpers package.

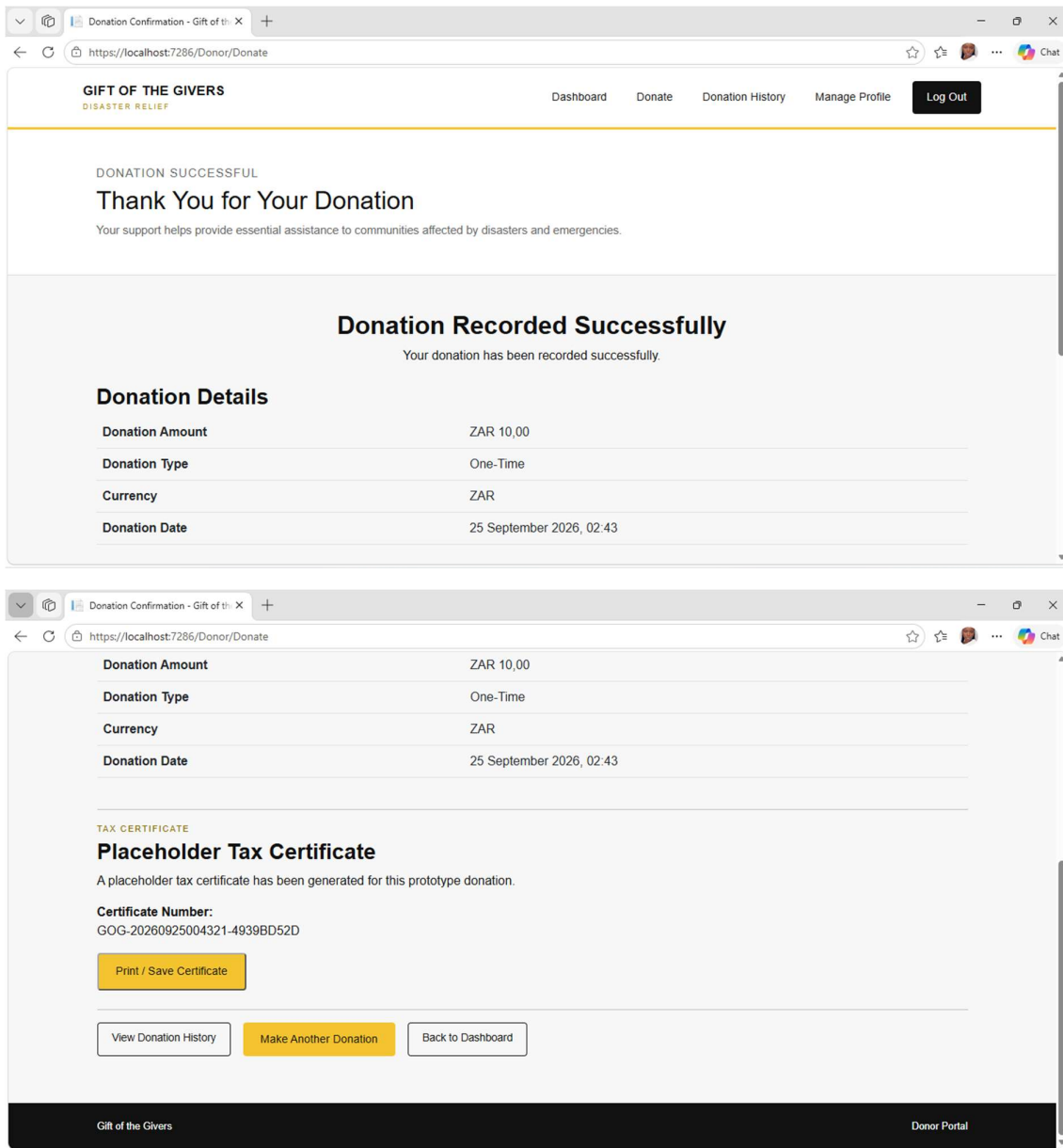
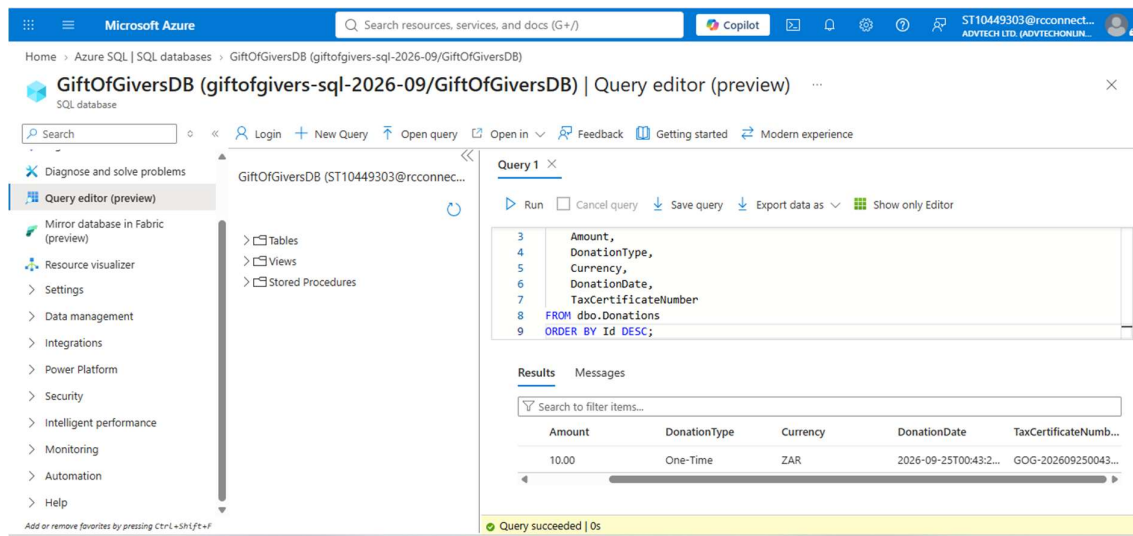
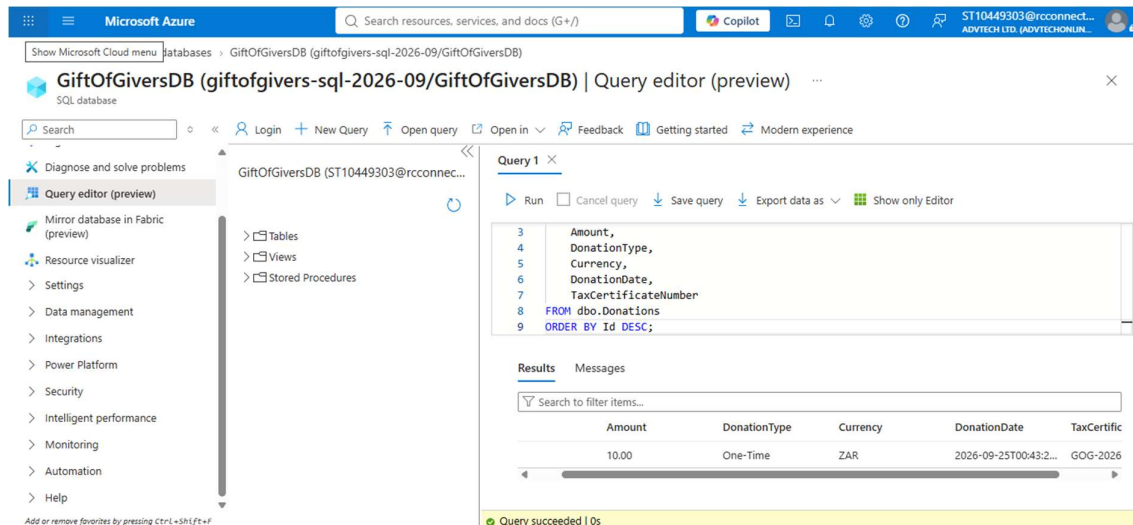


Figure 65: Azure SQL Query Editor showing the successful donation record stored in dbo.Donations.



Short group reflection

Thabelo

Worked on Azure Functions, Azure Artifacts, Azure SQL, Azure Pipelines, and integration of the helper package into the donation process.

Thubalakhe

Worked on the Home page and related user-interface functionality.

Nkateko

Worked on the About and information-related functionality.

Kgolagano

Worked on the Employee and Volunteer-related functionality.

Group reflection:

We divided the development work according to assigned features and collaborated using Azure Repos, branches, commits, and merges. The project gave the group practical experience in ASP.NET Core development, Azure cloud services, Git version control, CI pipelines, and NuGet package management.